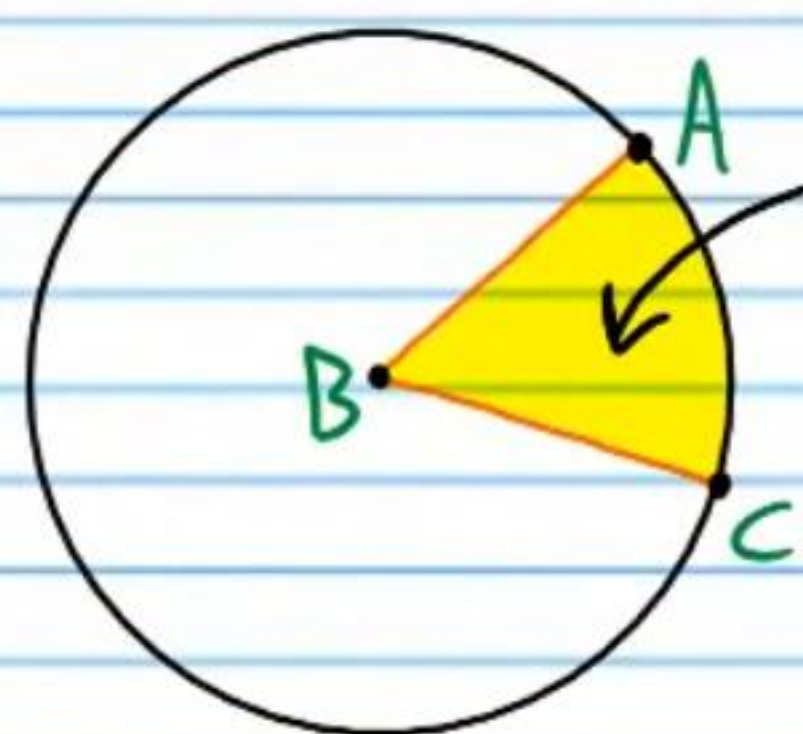


Circles (Part II): Sectors, Chords, and Graphs of Circles

Sector of a Circle

Example: Sector ABC below.



Sector: The region in a circle bounded by two radii and the corresponding arc.

Area of a Circle

$$A = \pi r^2$$

r is radius

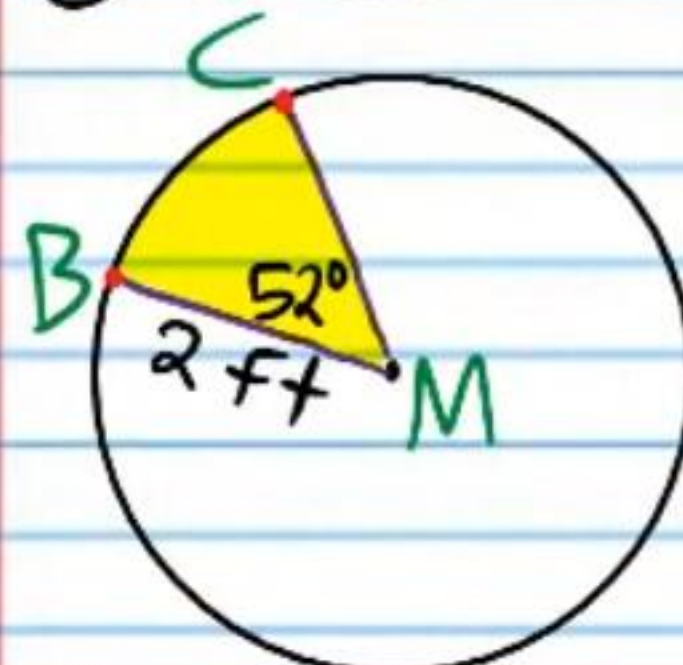
Area of a Sector

$$A = \frac{\theta}{360} (\pi r^2)$$

θ is central angle and r is radius

If M is the center of the following circles, find the area of the sectors below. **You must use the π button on your calculator.** Round to the nearest tenth.

① BMC



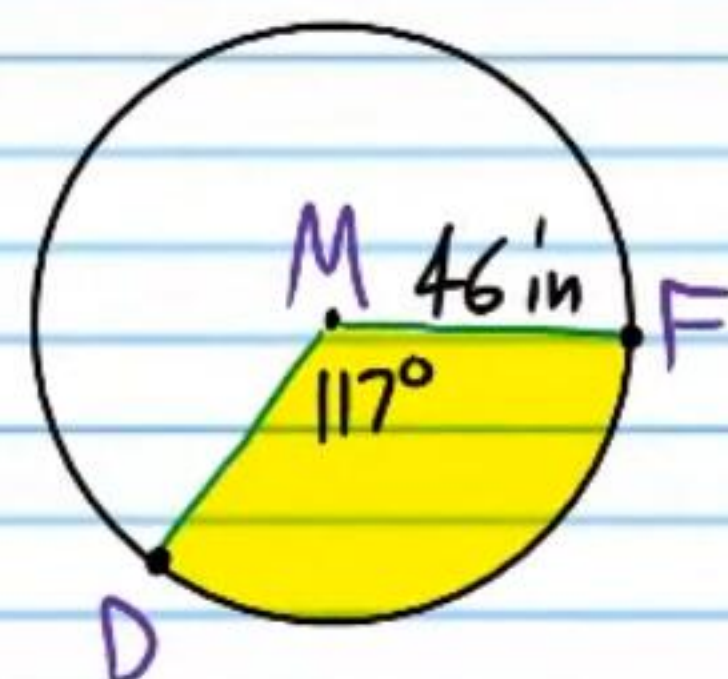
$$A = \frac{\theta}{360} (\pi r^2)$$

$$A = \frac{52}{360} (\pi (2)^2)$$

$$A = \frac{52}{360} (\pi (4))$$

$$A \approx 1.8 \text{ ft}^2$$

② DMF



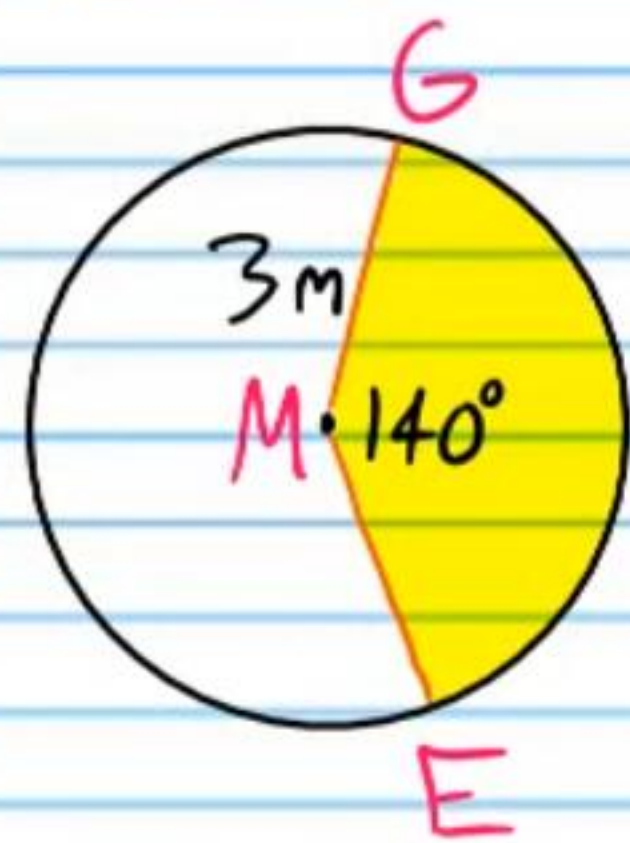
$$A = \frac{\theta}{360} (\pi r^2)$$

$$A = \frac{117}{360} (\pi (46)^2)$$

$$A = \frac{117}{360} (\pi (2,116))$$

$$A \approx 2,160.5 \text{ in}^2$$

③ EMG



$$A = \frac{\theta}{360} (\pi r^2)$$

$$A = \frac{140}{360} (\pi (3)^2)$$

$$A = \frac{140}{360} (\pi (9))$$

$$\boxed{A \approx 11 \text{ m}^2}$$

④ IMJ



$$A = \frac{\theta}{360} (\pi r^2)$$

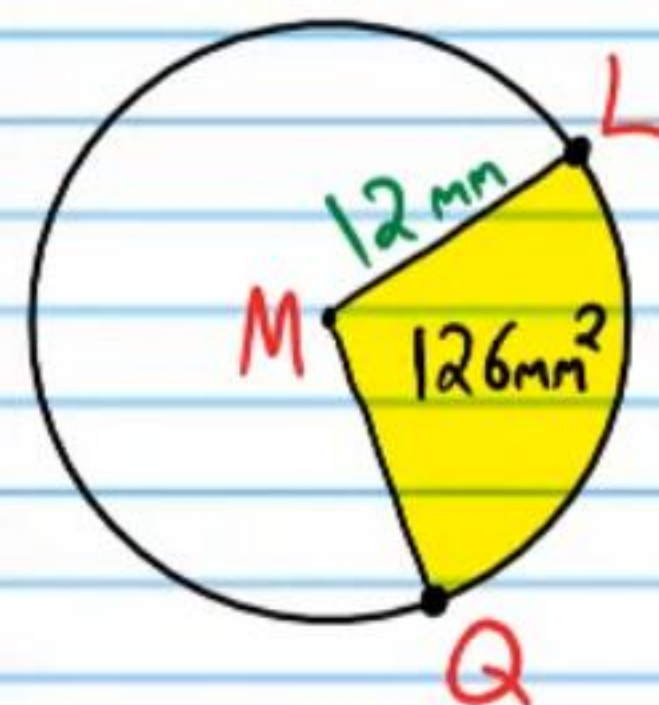
$$A = \frac{75}{360} (\pi (20)^2)$$

$$A = \frac{75}{360} (\pi (400))$$

$$\boxed{A \approx 261.8 \text{ in}^2}$$

If M is the center of the following circles, find the following values. You must use the π button on your calculator. Round to the nearest tenth.

⑤ $m\angle LMQ$



$$A = \frac{\theta}{360} (\pi r^2)$$

$$126 = \frac{\theta}{360} (\pi (12)^2)$$

$$126 = \frac{\theta}{360} (\pi (144))$$

$$(360)126 = \frac{\theta}{360} (144\pi) (360)$$

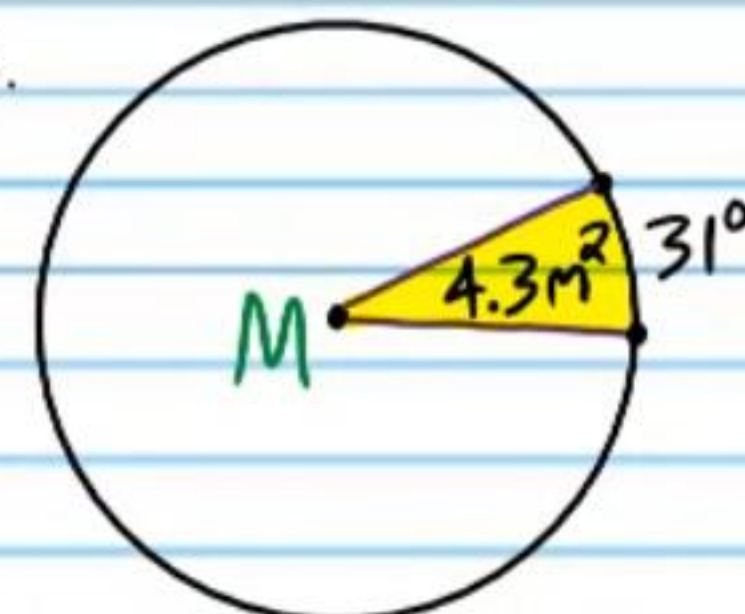
$$(360)(126) = \theta (144\pi)$$

$$\frac{(360)(126)}{144\pi} = \frac{\theta (144\pi)}{144\pi}$$

$$\frac{(360)(126)}{144\pi} = \theta$$

$$\boxed{\theta \approx 100.3^\circ}$$

⑥ Radius.



$$A = \frac{\theta}{360} (\pi r^2)$$

$$4.3 = \frac{31}{360} (\pi r^2)$$

$$(360)4.3 = \frac{31}{360} (\pi r^2) (360)$$

$$(360)(4.3) = 31 (\pi r^2)$$

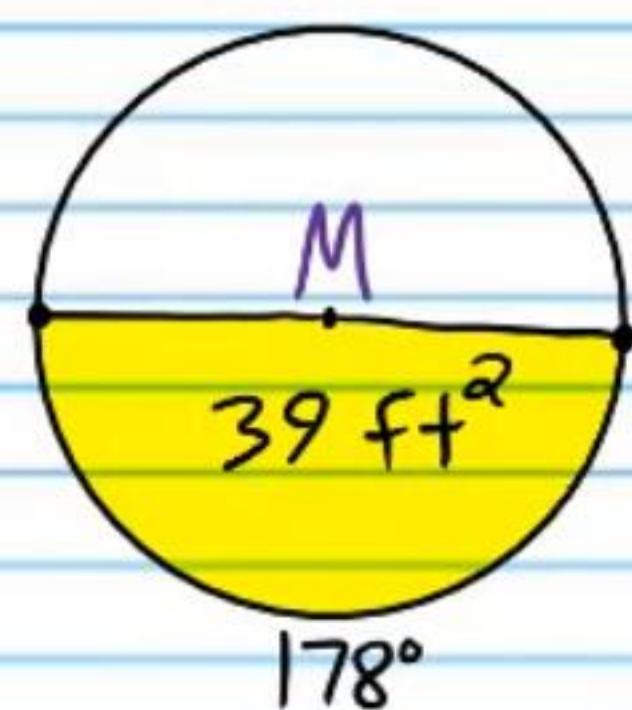
$$\frac{(360)(4.3)}{31\pi} = \frac{(31\pi) r^2}{31\pi}$$

$$\frac{(360)(4.3)}{31\pi} = r^2$$

$$\pm \sqrt{\frac{(360)(4.3)}{31\pi}} = r$$

$$\boxed{4 \text{ m} \approx r}$$

⑦ Area of $\odot M$.



$$A = \frac{\theta}{360} (\pi r^2)$$

$$39 = \frac{178}{360} (\pi r^2)$$

$$(360)39 = \frac{178}{360} (\pi r^2)(360)$$

$$(360)(39) = 178 (\pi r^2)$$

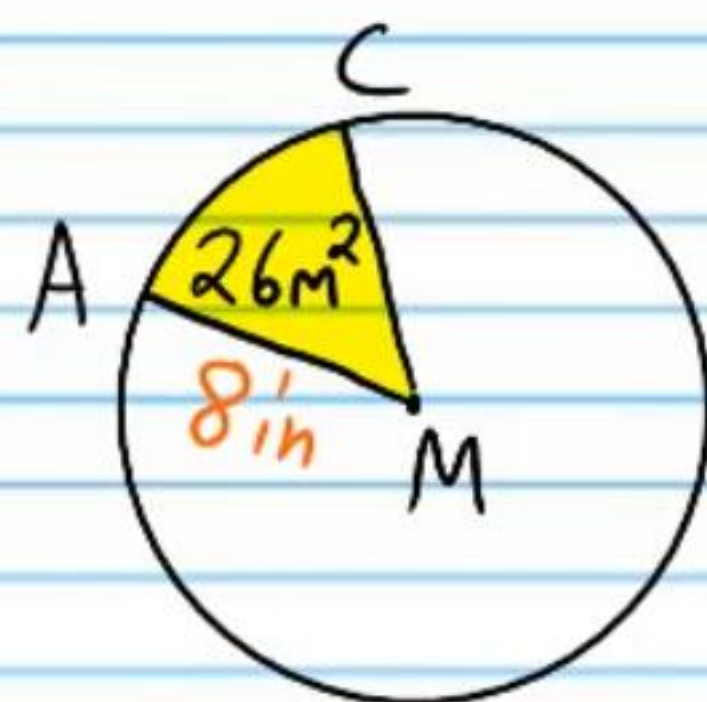
$$(360)(39) = 178 (\pi r^2)$$

$$\frac{178}{178} \quad \frac{178}{178}$$

$$\frac{(360)(39)}{178} = \pi r^2$$

$$78.9 \text{ ft}^2 \approx \pi r^2$$

⑧ $m\angle AMC$



$$A = \frac{\theta}{360} (\pi r^2)$$

$$26 = \frac{\theta}{360} (\pi (8)^2)$$

$$26 = \frac{\theta}{360} (\pi (64))$$

$$(360)26 = \frac{\theta}{360} (64\pi)(360)$$

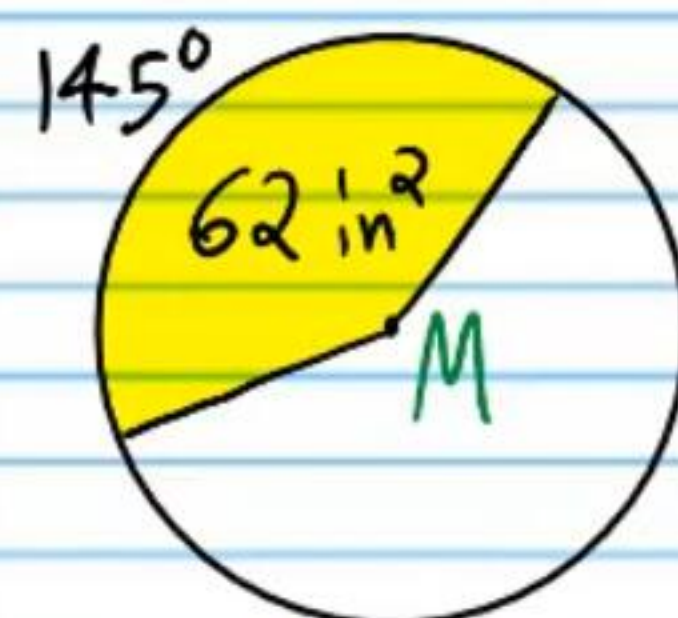
$$(360)(26) = \theta(64\pi)$$

$$\frac{(360)(26)}{64\pi} = \frac{\theta(64\pi)}{64\pi}$$

$$\frac{(360)(26)}{64\pi} = \theta$$

$$46.6^\circ \approx \theta$$

⑨ Diameter



$$A = \frac{\theta}{360} (\pi r^2)$$

$$62 = \frac{145}{360} (\pi r^2)$$

$$(360)62 = \frac{145}{360} (\pi r^2)(360)$$

$$(360)(62) = 145 (\pi r^2)$$

$$\frac{(360)(62)}{145\pi} = \frac{145\pi r^2}{145\pi}$$

$$\frac{(360)(62)}{145\pi} = r^2$$

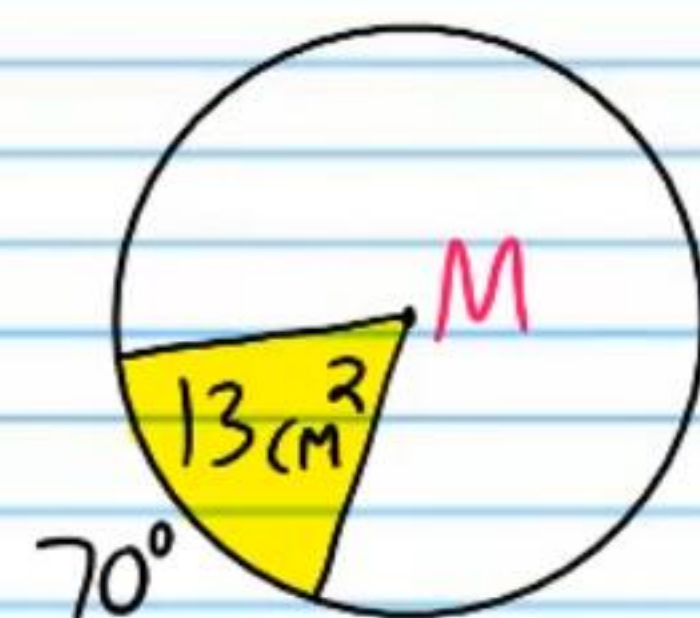
$$\pm \sqrt{\frac{(360)(62)}{145\pi}} = r$$

$$6.9998 \approx r$$

$$2(6.9998) \approx r$$

$$14 \text{ in} \approx D$$

⑩ Area of $\odot M$



$$A = \frac{\theta}{360} (\pi r^2)$$

$$13 = \frac{70}{360} (\pi r^2)$$

$$(360)13 = \frac{70}{360} (\pi r^2)(360)$$

$$(360)(13) = 70 (\pi r^2)$$

$$\frac{(360)(13)}{70} = \frac{70 (\pi r^2)}{70}$$

$$\frac{(360)(13)}{70} = \pi r^2$$

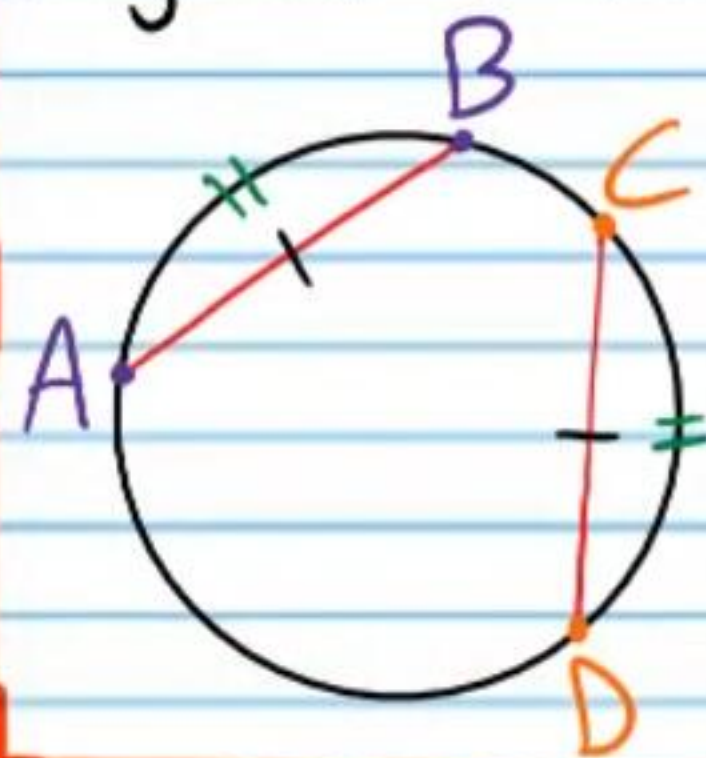
$$66.9 \text{ cm}^2 \approx \pi r^2$$

Chords

Congruent Arcs: Arcs with equal measures (must be in the same circle or congruent circles).

Congruent Circles: Circles with congruent radii.

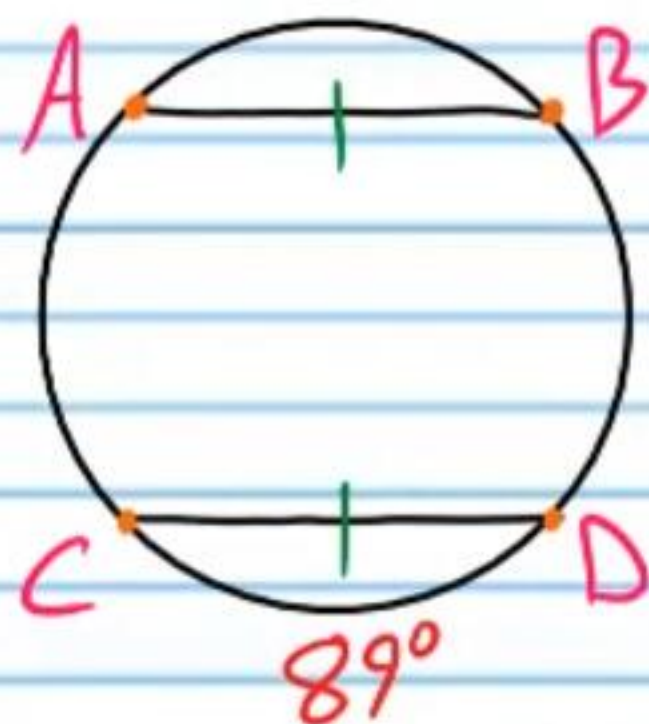
Arc-Chord Pair Congruence Theorem: The minor arcs created by two chords in the same circle are congruent if and only if the chords are congruent.



$\widehat{AB} \cong \widehat{CD}$ if and only if $\overline{AB} \cong \overline{CD}$.

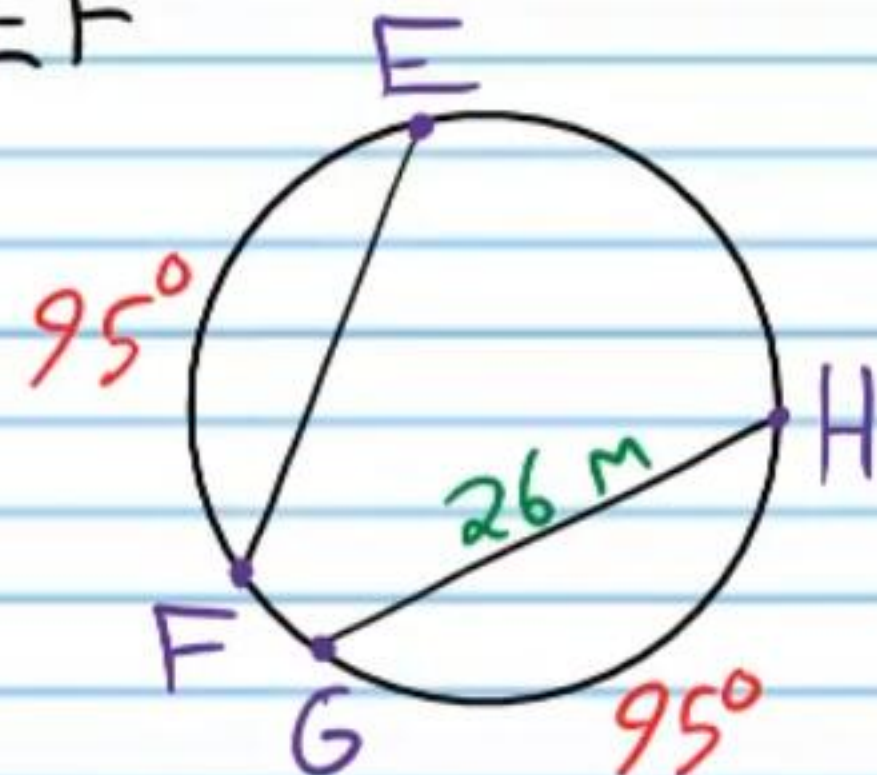
Find the following values.

⑪ $m\widehat{AB}$



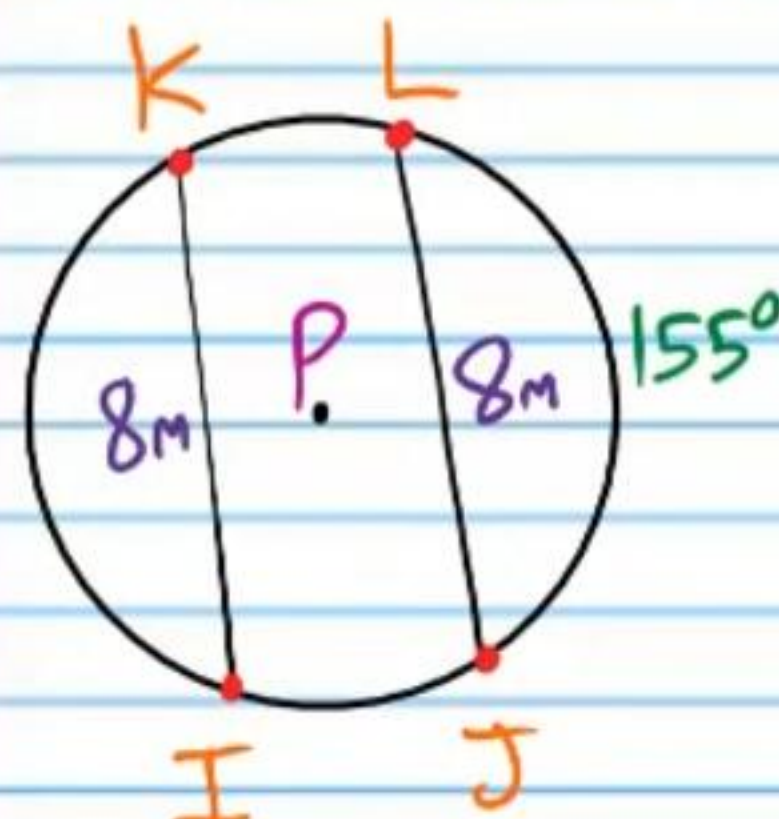
$$m\widehat{AB} = 89^\circ$$

⑫ EF



$$EF = 26m$$

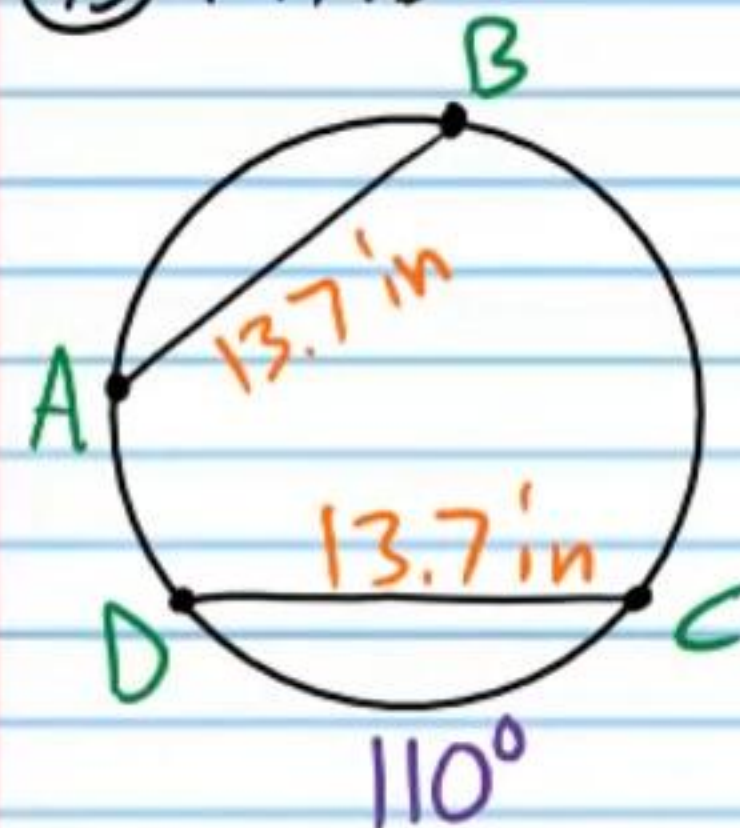
⑬ $m\angle IPK$ if P is the center of the circle.



$$m\widehat{KI} = 155^\circ = m\angle IPK$$

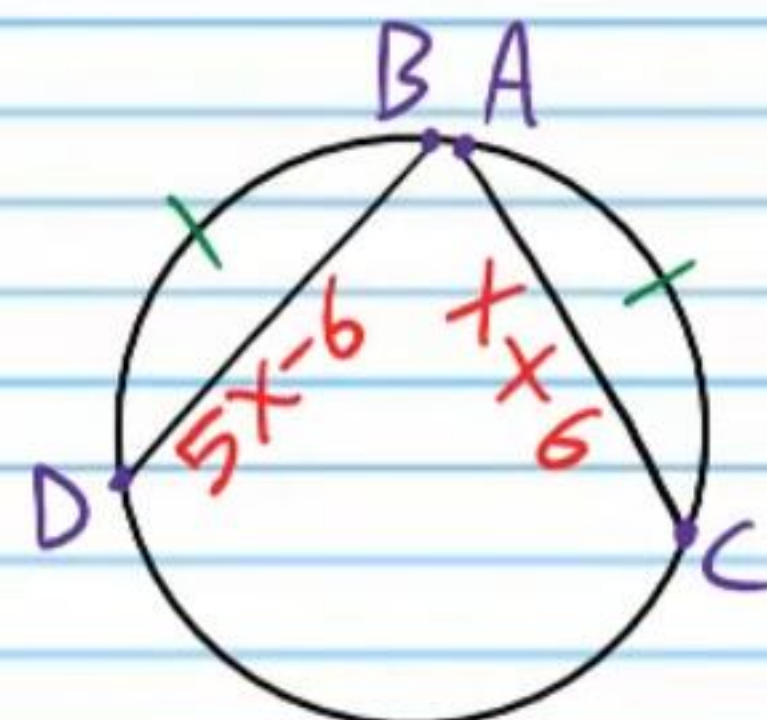
$$m\angle IPK = 155^\circ$$

⑮ $m\widehat{AB}$



$$m\widehat{AB} = 110^\circ$$

⑭ AC

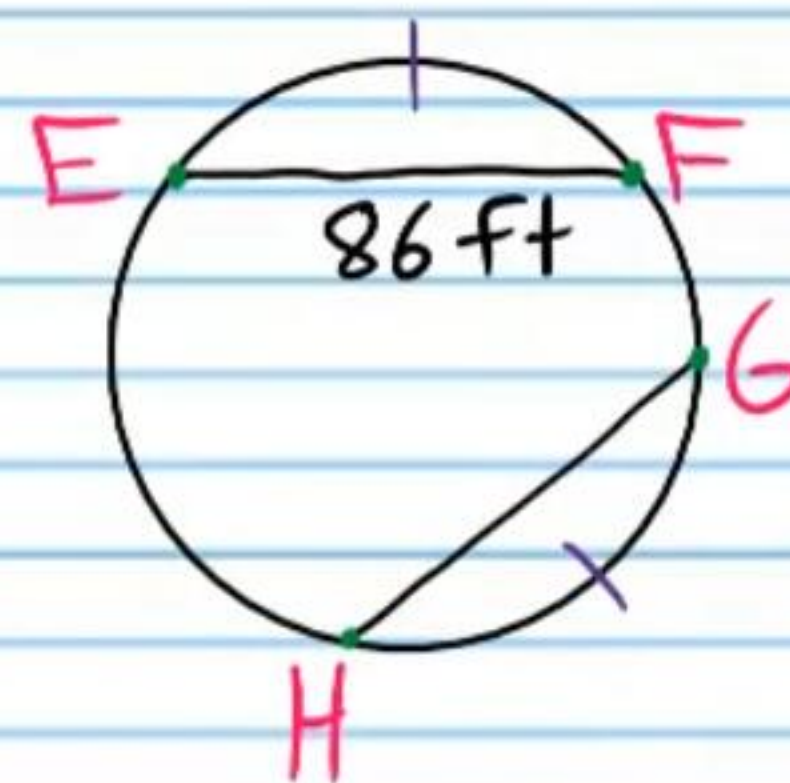


$$\begin{aligned} 5x-6 &= x+6 \\ 4x &= 12 \\ x &= 3 \end{aligned}$$

$$AC = 3+6$$

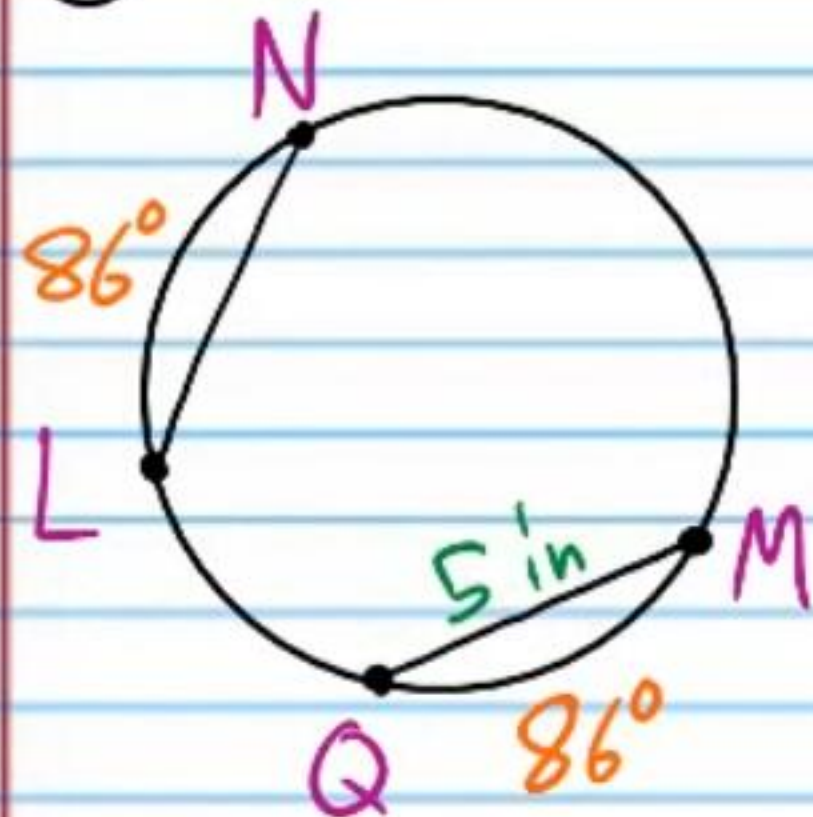
$$AC = 9$$

⑯ HG



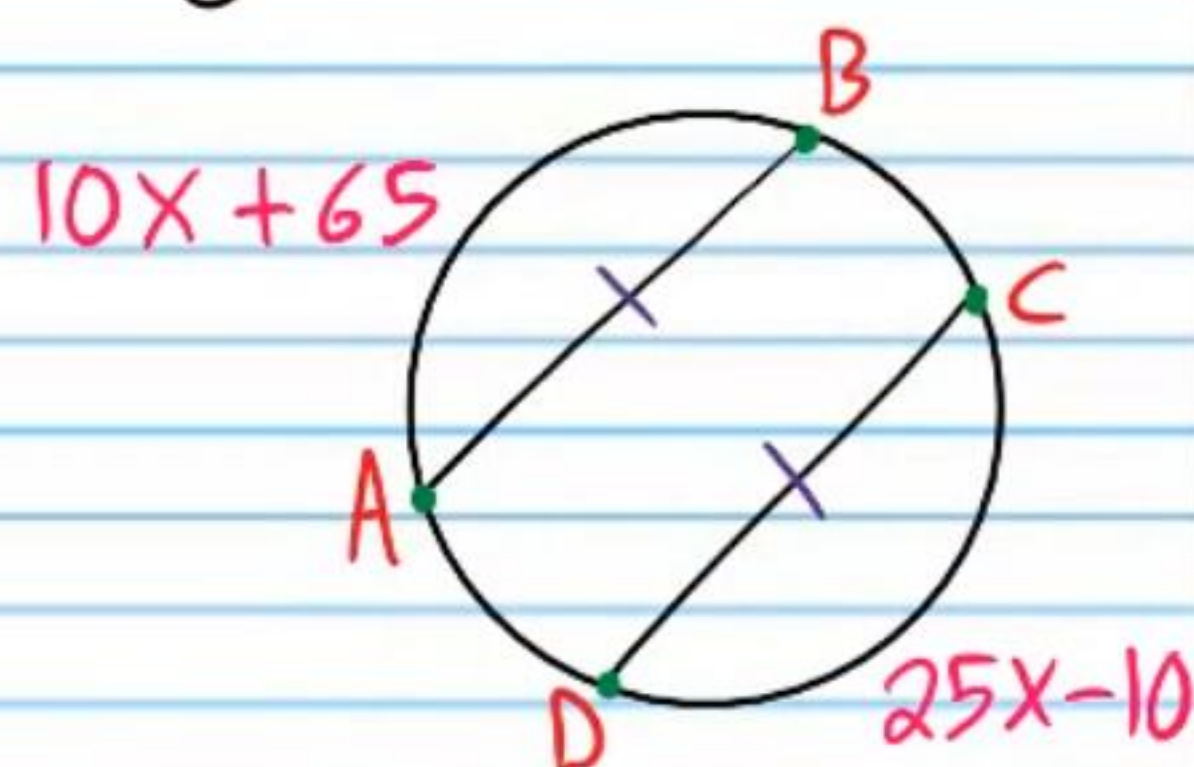
$$HG = 86ft$$

⑪ LN



$$\boxed{LN = 5 \text{ in}}$$

⑫ $m\widehat{DC}$

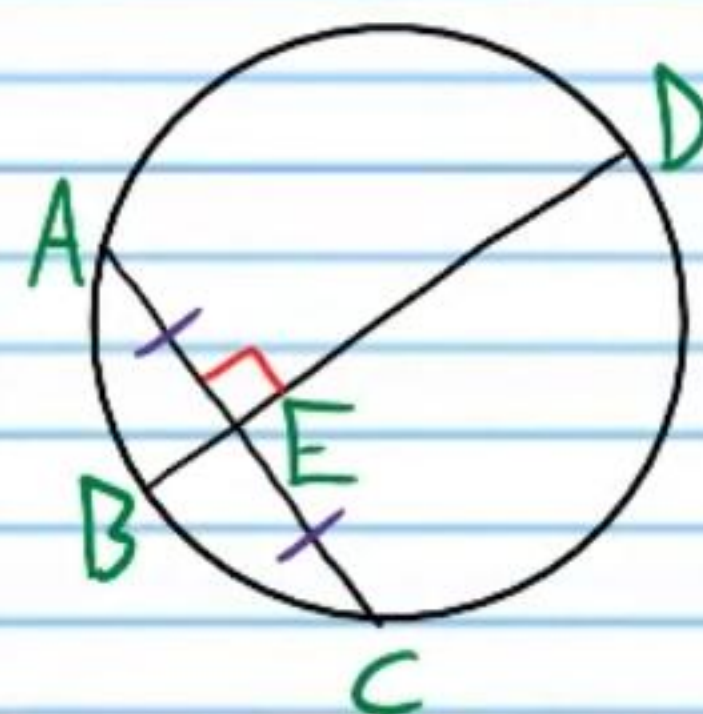


$$\begin{aligned} 10X + 65 &= 25X - 10 \\ 75 &= 15X \\ 5 &= X \end{aligned}$$

$$m\widehat{DC} = 25(5) - 10 = 115^\circ$$

$$\boxed{m\widehat{DC} = 115^\circ}$$

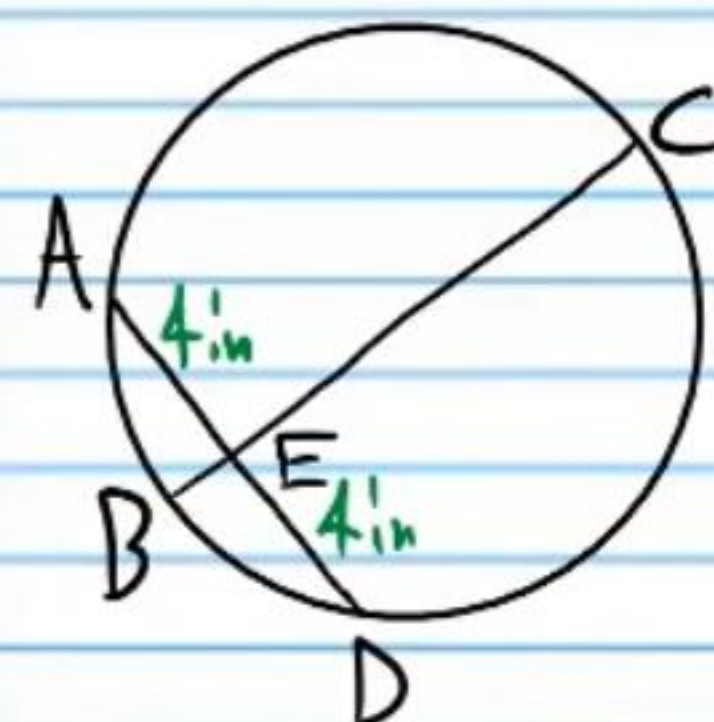
Chord Perpendicular Bisector Theorem: If a chord in a circle is a perpendicular bisector of another chord, then the first chord must be a diameter of the circle.



If $\overline{BD} \perp \overline{AC}$ and $AE = EC$, then $\overline{BD}$ is a diameter of $\odot F$.

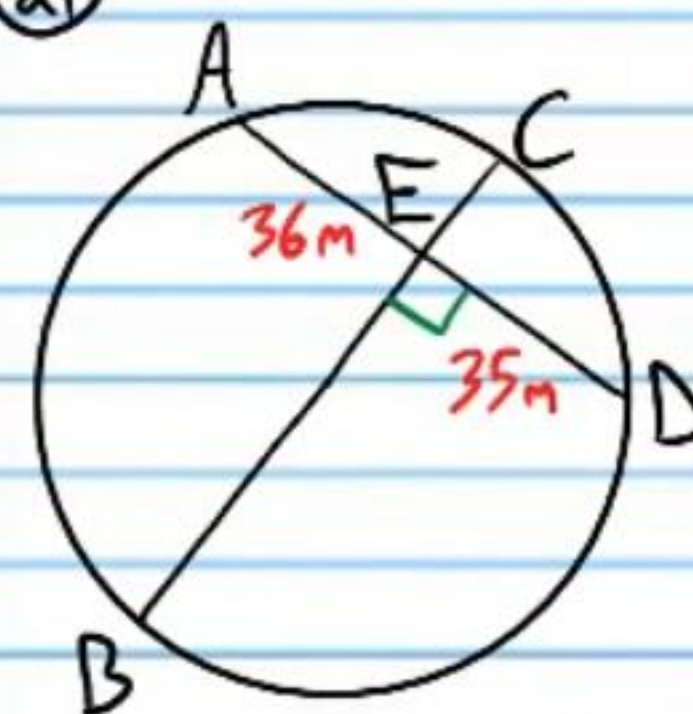
Is $\overline{BC}$ a diameter? Explain.

⑬ $m\angle DEC = 90^\circ$



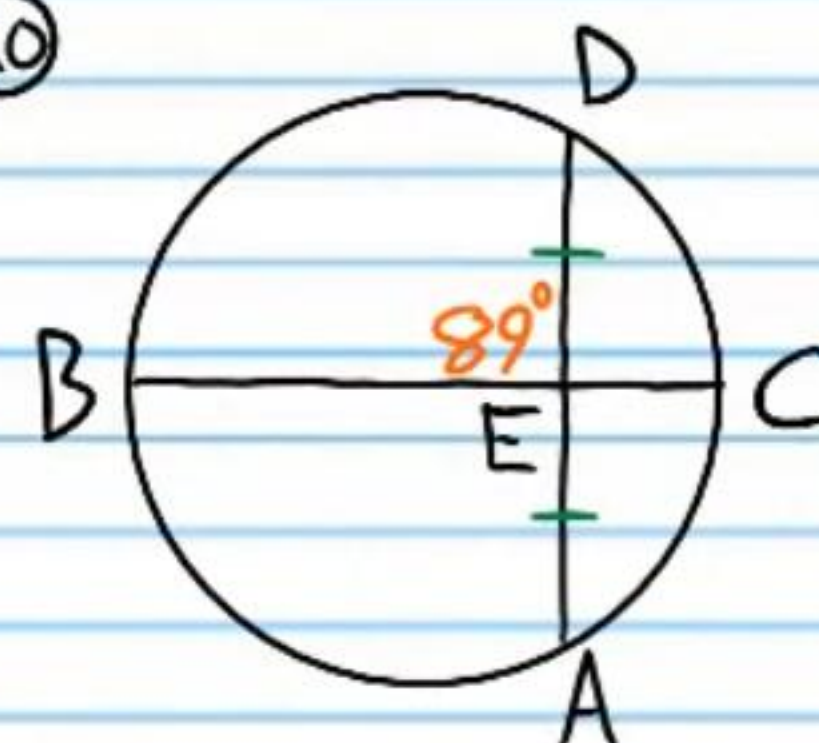
$\overline{BC}$ is a diameter because it's a $\perp$ bisector of chord $\overline{AD}$.

⑭



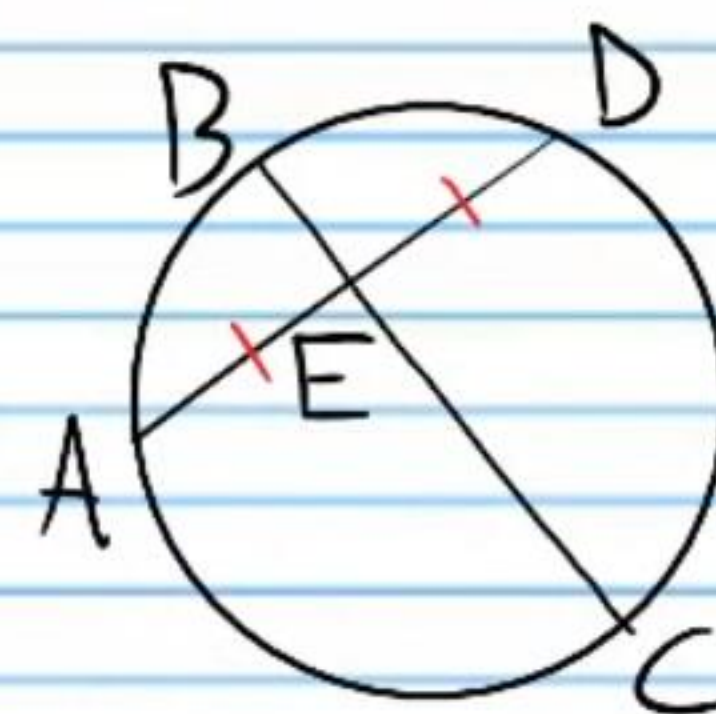
We cannot use the Chord $\perp$ Bisector Thm b/c $\overline{BC}$ does not bisect $\overline{AD}$.

⑮



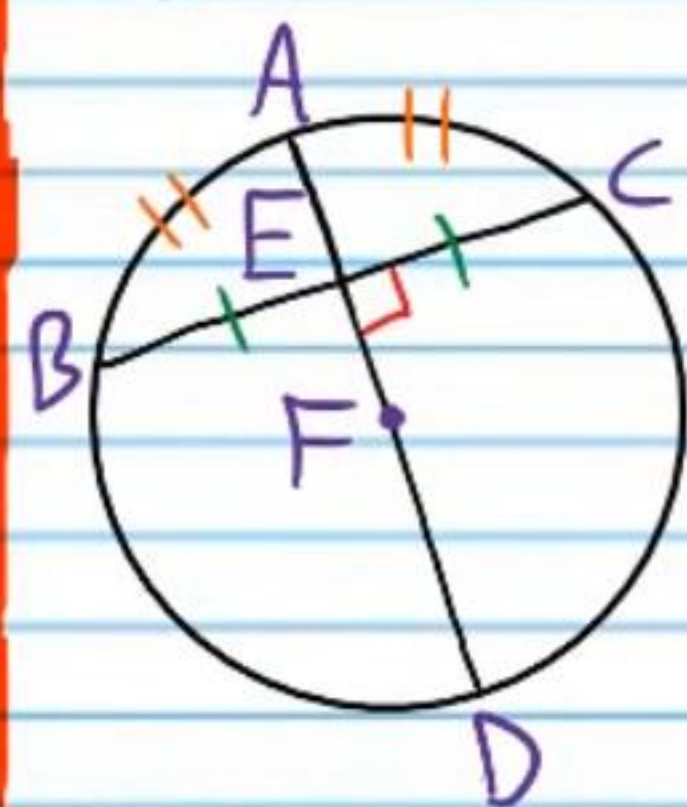
We cannot use the Chord $\perp$ Bisector Thm b/c $\overline{BC}$ is not $\perp$ to $\overline{DA}$.

⑯ $\overline{BC} \perp \overline{AD}$



$\overline{BC}$ is a diameter b/c it's a $\perp$ bisector of $\overline{AD}$.

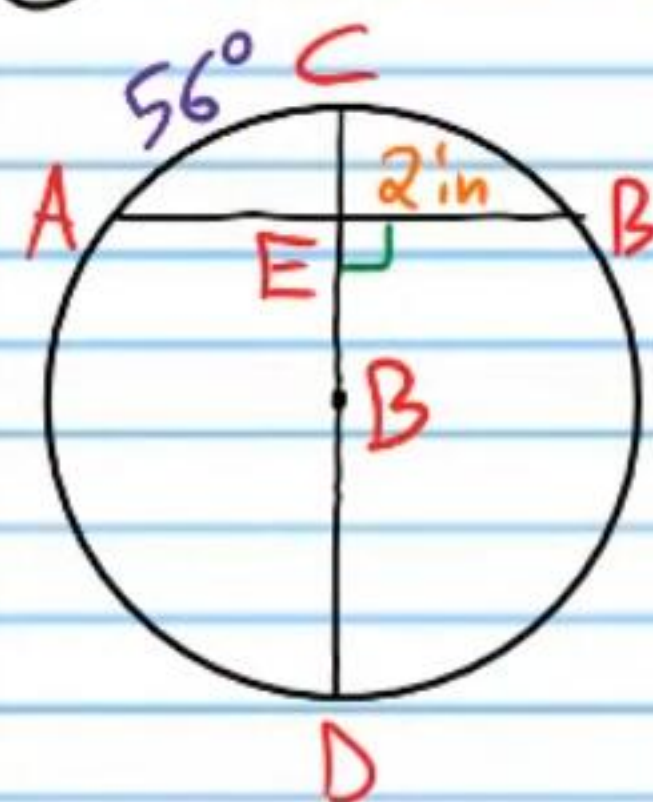
Perpendicular Diameter Theorem: If a diameter or a radius of a circle is perpendicular to a chord, then it bisects the chord and its arc.



If $\overline{AD}$ is a diameter of $\odot F$ and $\overline{AD} \perp \overline{BC}$, then $BE = EC$ and $m\widehat{AB} = m\widehat{AC}$.

If B is the center of each circle below, find the following values.

(23) $m\widehat{CB}$ & AB.

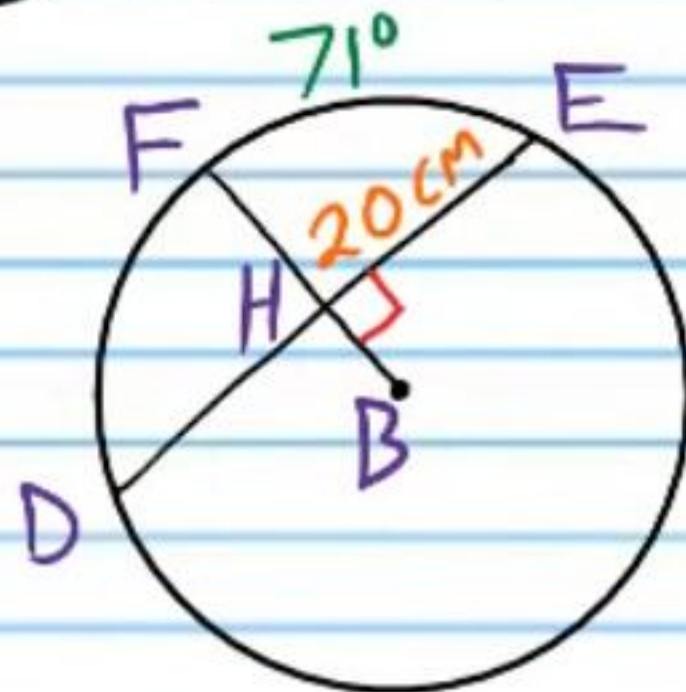


$$m\widehat{CB} = 56^\circ$$

$$AB = 2(2) = 4$$

$$AB = 4 \text{ in}$$

(24) DH & $m\widehat{DE}$

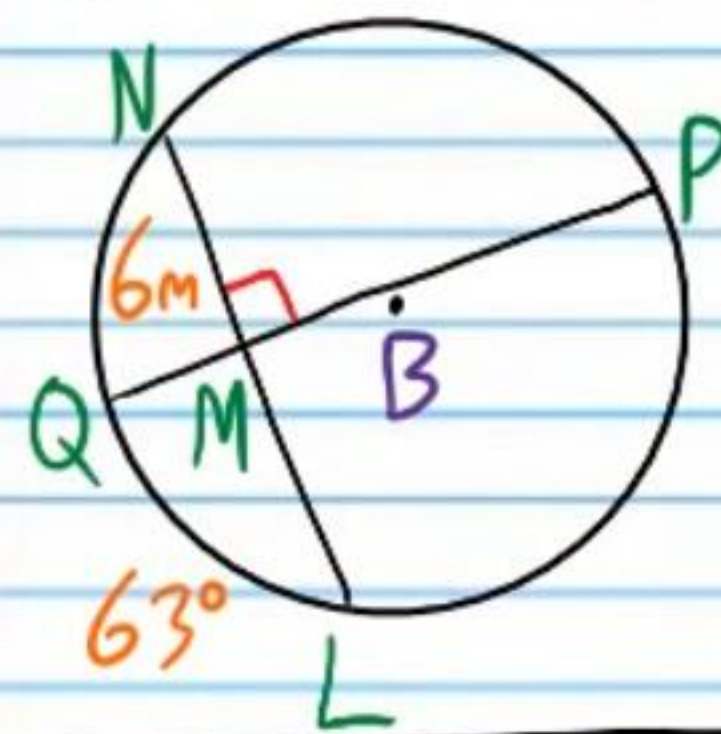


$$DH = 20 \text{ cm}$$

$$m\widehat{DE} = 2(71^\circ) = 142^\circ$$

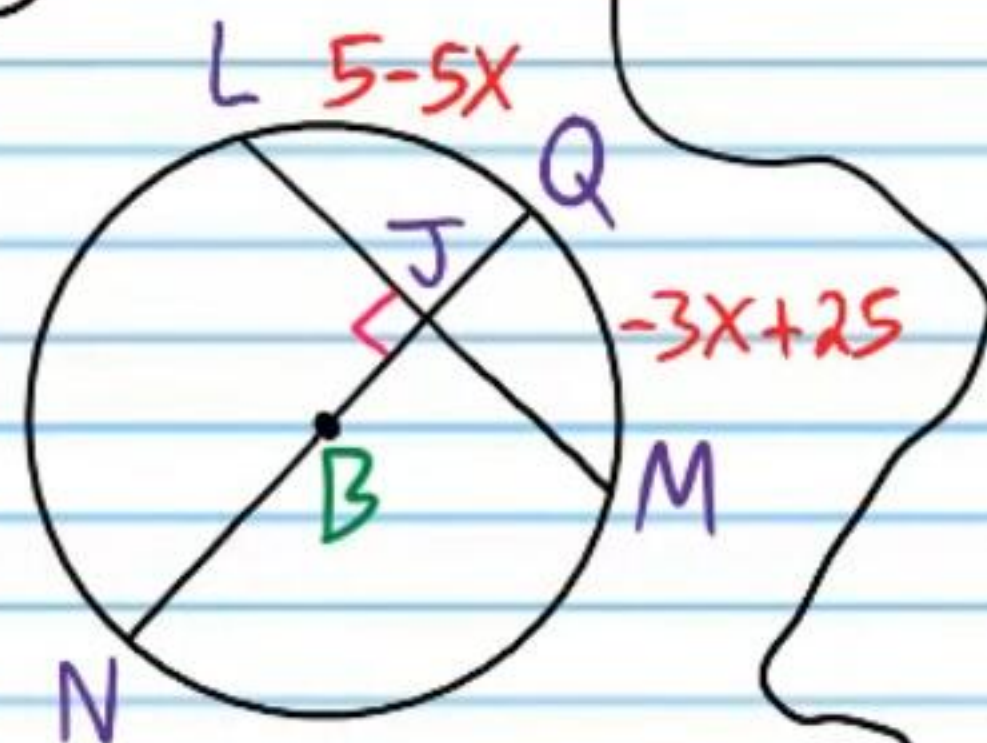
$$m\widehat{DE} = 142^\circ$$

(25) NL & $m\widehat{NL}$



We cannot use the $\perp$ Diameter Thm b/c QP is not a diameter.

(27) $m\widehat{LNM}$



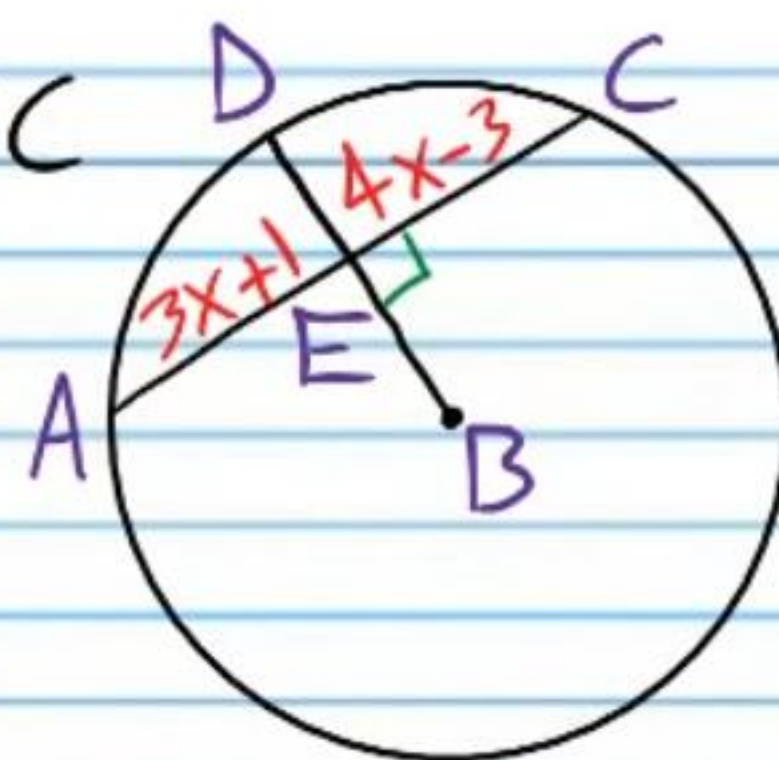
$$\begin{aligned} 5 - 5x &= -3x + 25 \\ -20 &= 2x \\ -10 &= x \end{aligned}$$

$$m\widehat{LQ} = 5 - 5(-10) = 55^\circ$$

$$m\widehat{LM} = 2(55) = 110^\circ$$

$$m\widehat{LNM} = 360 - 110 = 250^\circ$$

(26) EC

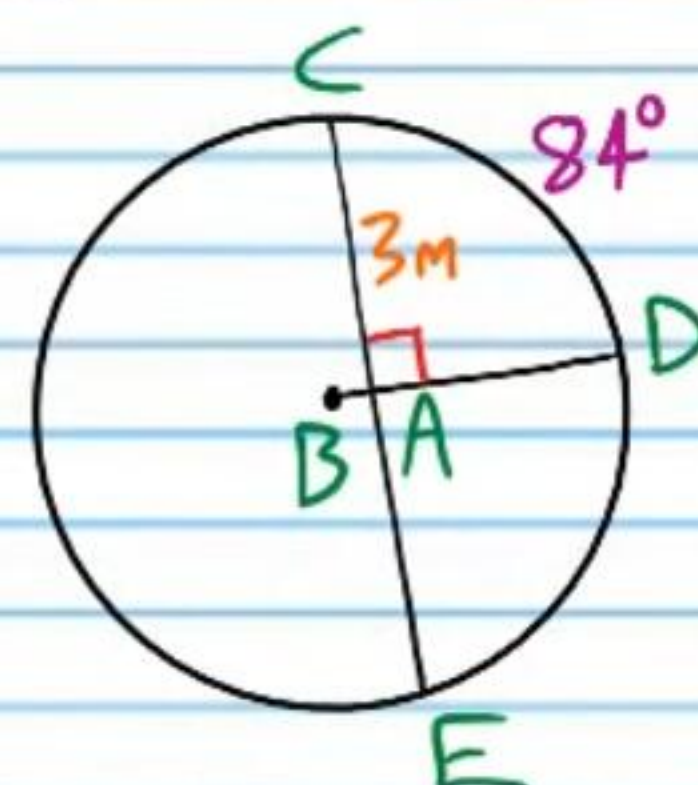


$$\begin{aligned} 3x + 1 &= 4x - 3 \\ 4 &= x \end{aligned}$$

$$EC = 4(4) - 3$$

$$EC = 13$$

(28) AE & $m\widehat{CE}$

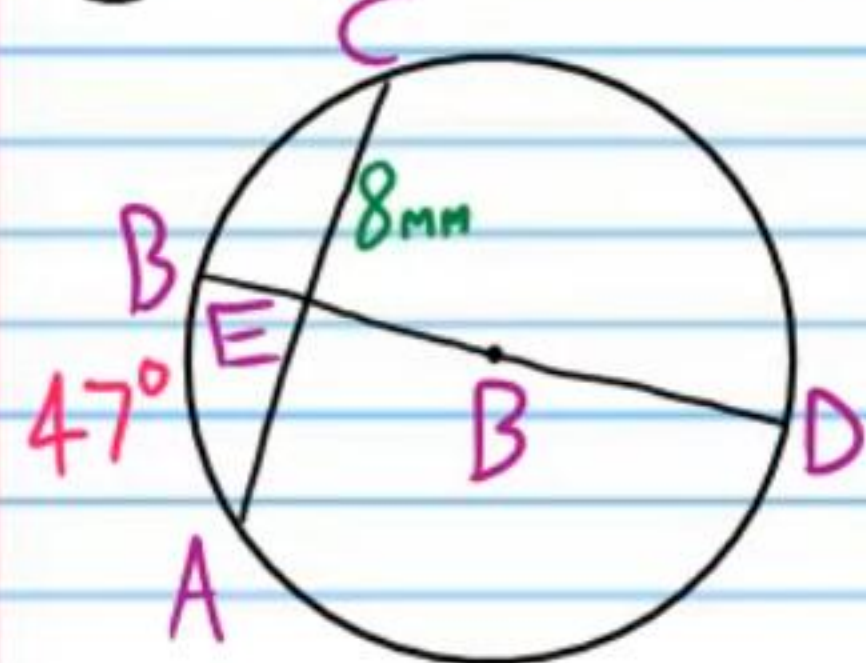


$$AE = 3m$$

$$m\widehat{CE} = 2(84)$$

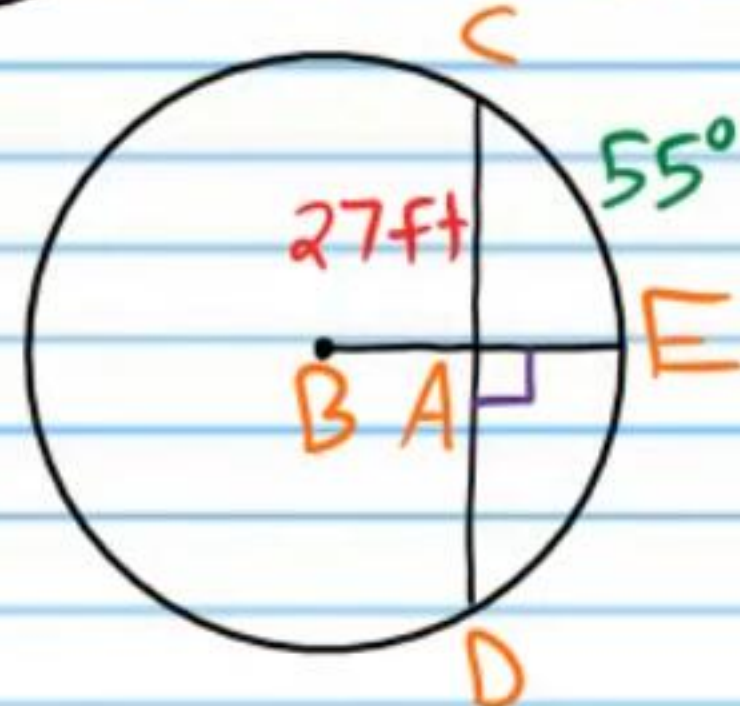
$$m\widehat{CE} = 168^\circ$$

29) $m\widehat{BC}$ & AC



We cannot use the $\perp$ Diameter Thm b/c BD might not be $\perp$ to AC.

30) $m\widehat{CD}$ & AD

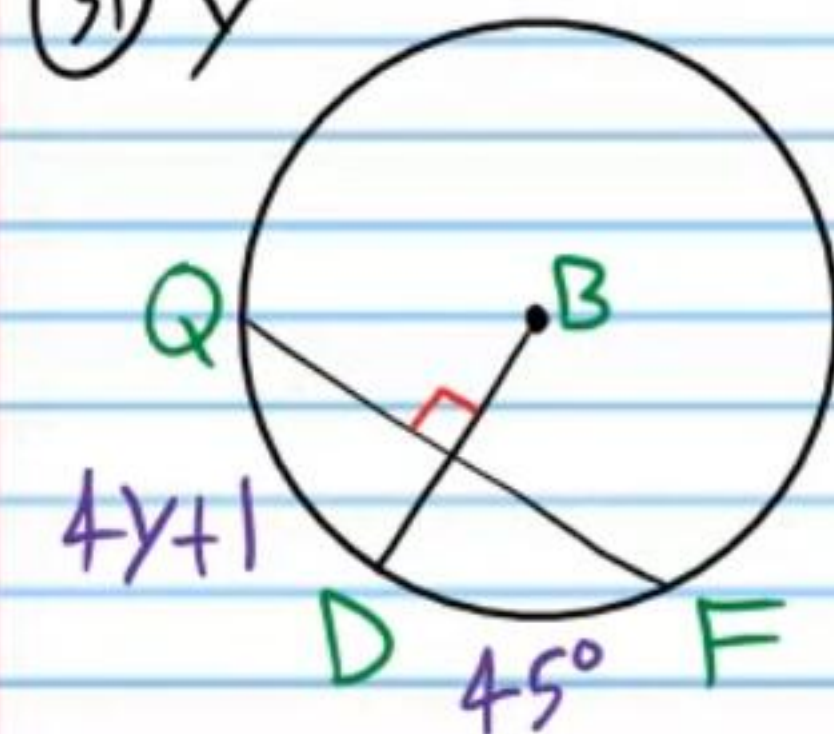


$$AD = 27ft$$

$$m\widehat{CD} = 2(55) = 110^\circ$$

$$m\widehat{CD} = 110^\circ$$

31) y

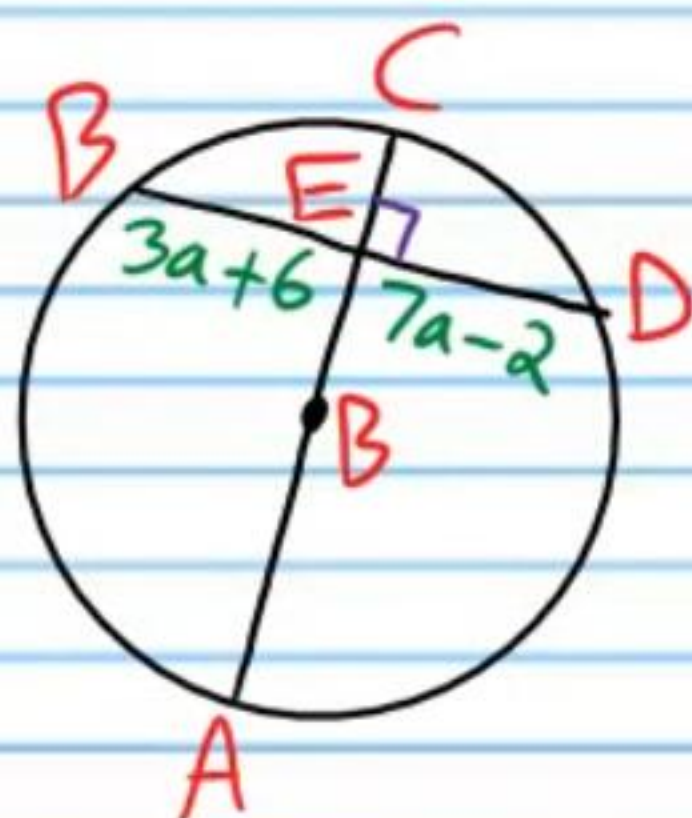


$$4y+1 = 45$$

$$4y = 44$$

$$y = 11$$

32) BD



$$3a+6 = 7a-2$$

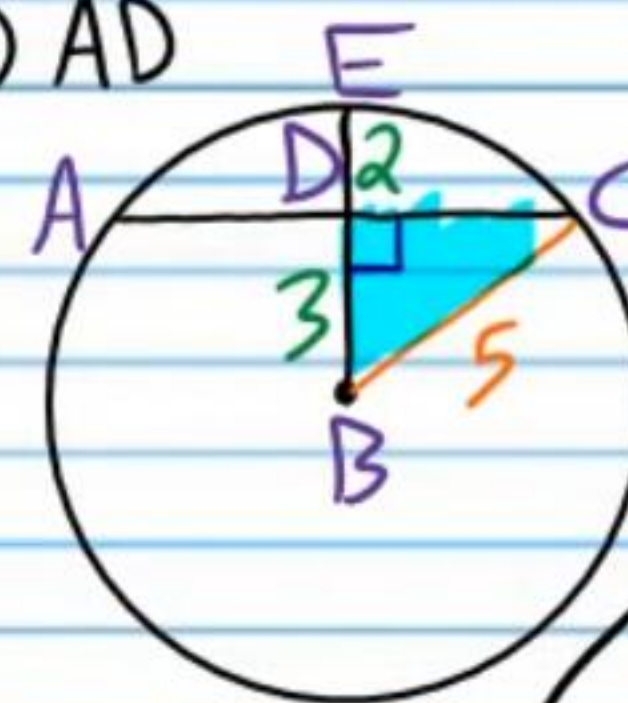
$$8 = 4a$$

$$2 = a$$

$$BE = 3(2)+6 = 12$$

$$BD = 2(12) = 24$$

33) AD



$$a^2 + b^2 = c^2$$

$$3^2 + (DC)^2 = 5^2$$

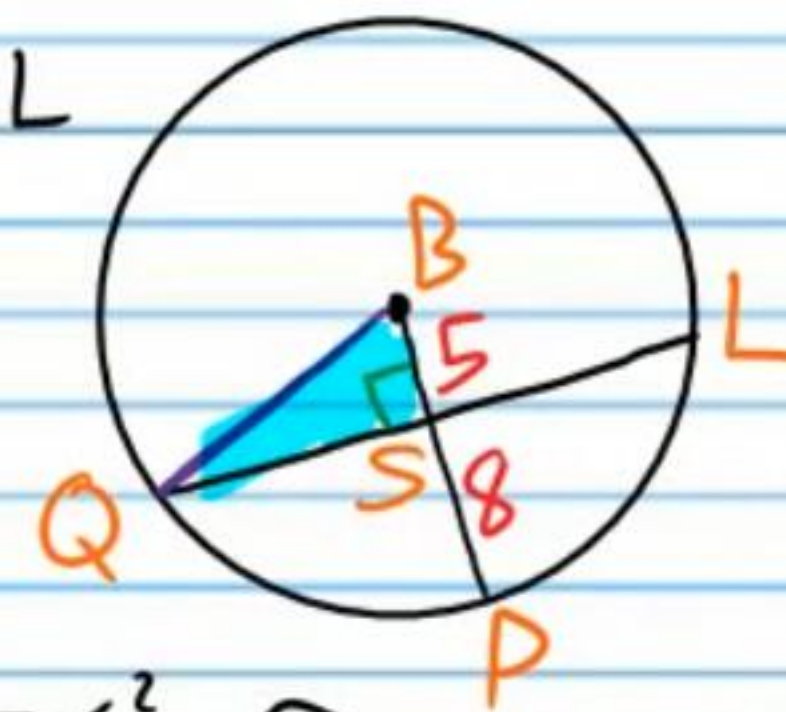
$$9 + (DC)^2 = 25$$

$$(DC)^2 = 16$$

$$DC = 4$$

$$AD = 4$$

34) QL



$$a^2 + b^2 = c^2$$

$$5^2 + (QS)^2 = 13^2$$

$$25 + (QS)^2 = 169$$

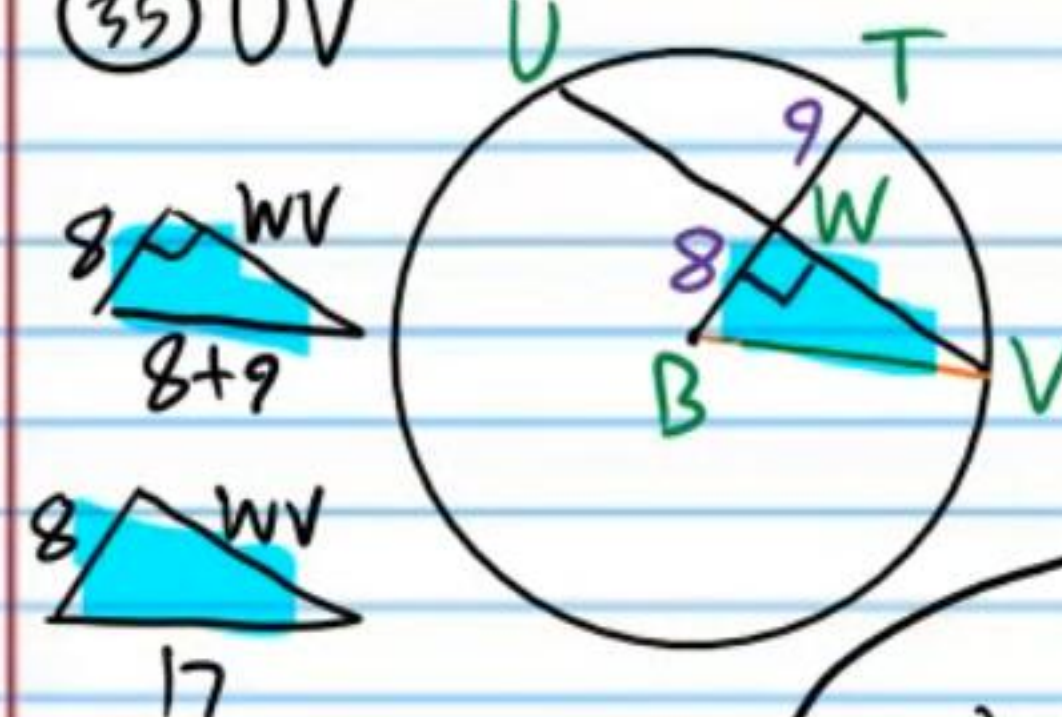
$$(QS)^2 = 144$$

$$QS = 12$$

$$QL = 2(12)$$

$$QL = 24$$

35) UV



$$a^2 + b^2 = c^2$$

$$8^2 + (WV)^2 = 17^2$$

$$64 + (WV)^2 = 289$$

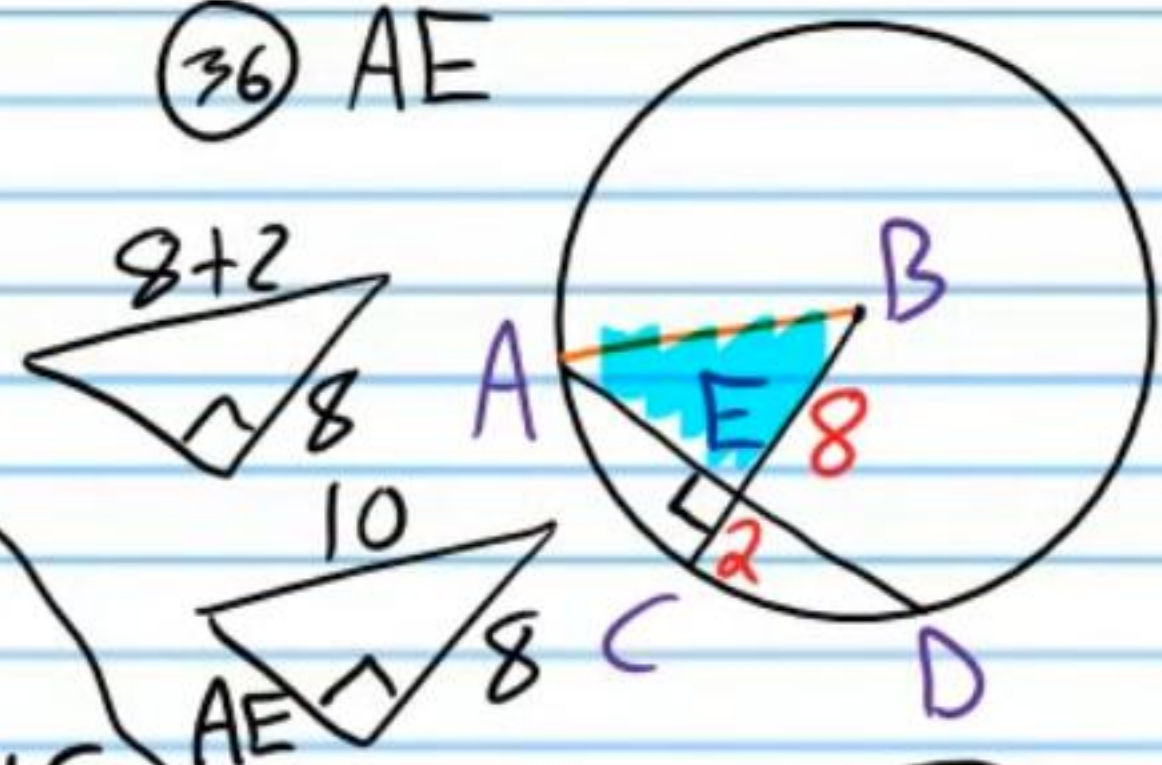
$$(WV)^2 = 225$$

$$WV = 15$$

$$UV = 2(15)$$

$$UV = 30$$

36) AE



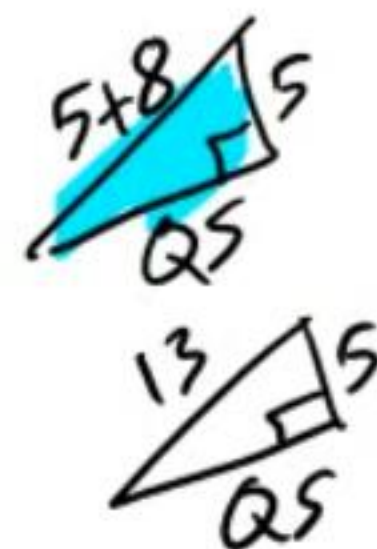
$$a^2 + b^2 = c^2$$

$$8^2 + (AE)^2 = 10^2$$

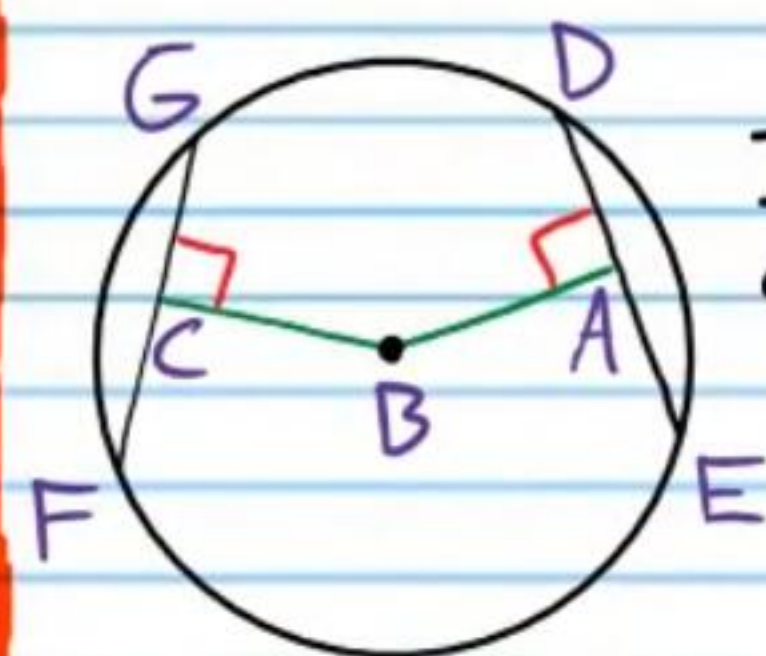
$$64 + (AE)^2 = 100$$

$$(AE)^2 = 36$$

$$AE = 6$$



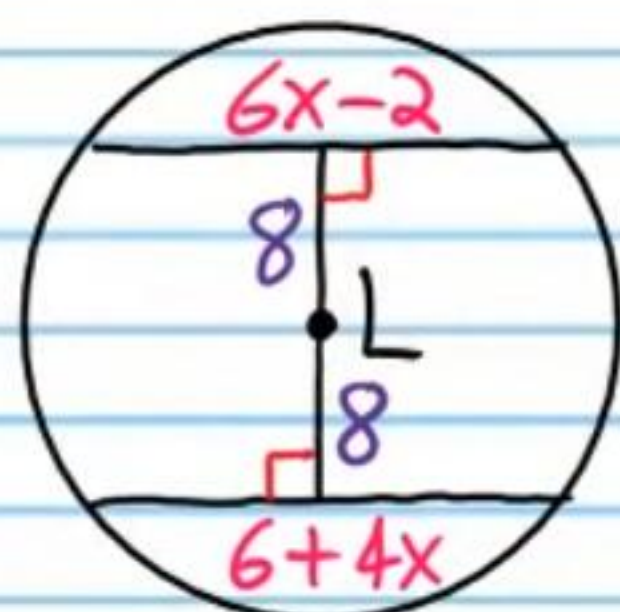
Equidistant Chords Theorem: Two chords in a circle are congruent if and only if they are equidistant from the center of the circle.



In $\odot B$, $\overline{DE} \cong \overline{GF}$ if and only if $\overline{CB} \cong \overline{BA}$.

If L is the center of each circle below, find the following values.

(37) X

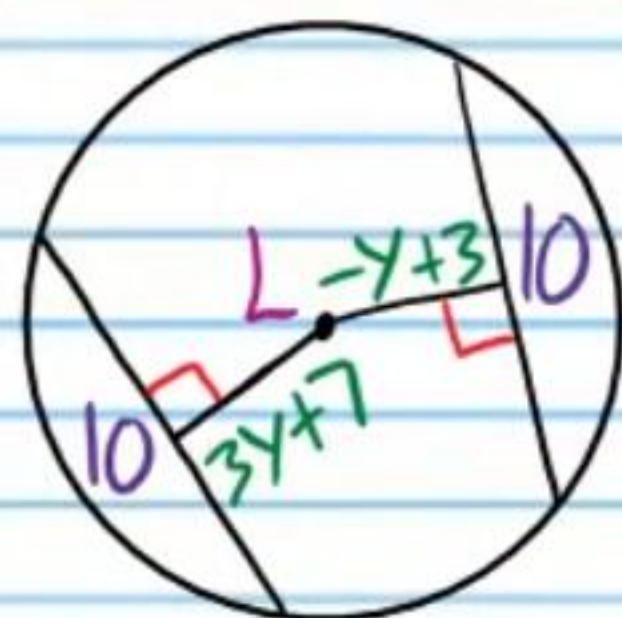


$$6x - 2 = 6 + 4x$$

$$2x = 8$$

$$\boxed{x = 4}$$

(38) y

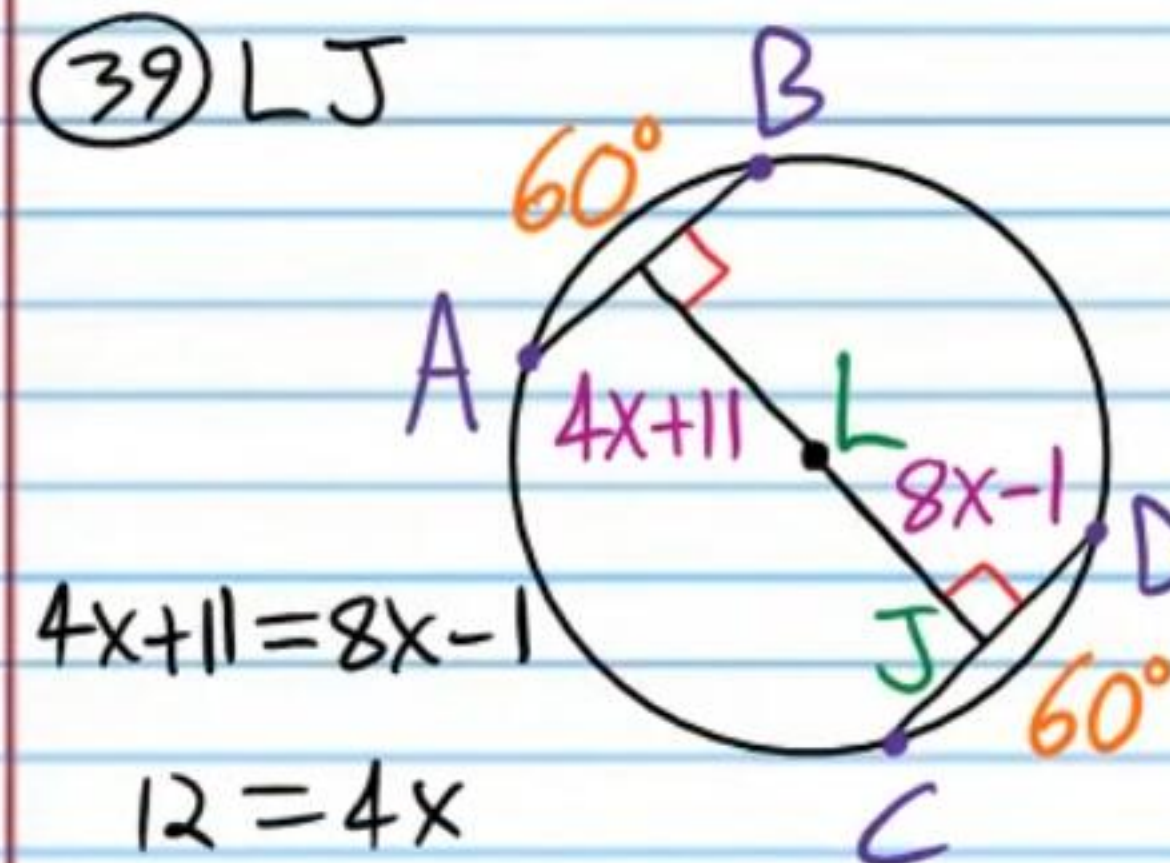


$$3y + 7 = -y + 3$$

$$4y = -4$$

$$\boxed{y = -1}$$

(39) LJ



$$4x + 11 = 8x - 1$$

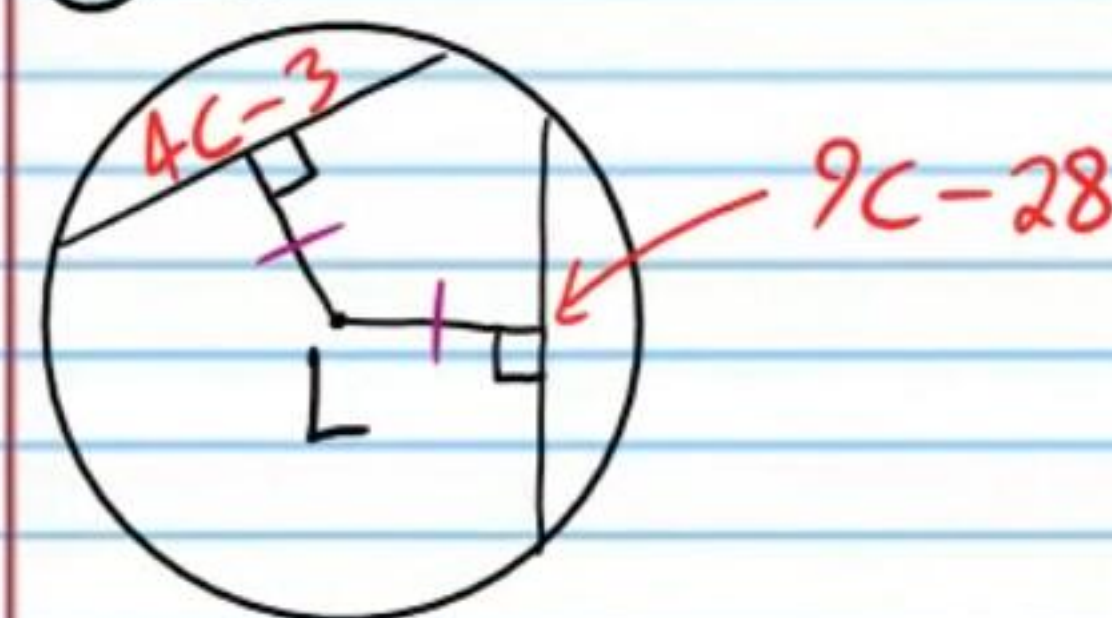
$$12 = 4x$$

$$3 = x$$

$$LJ = 8(3) - 1 = 23$$

$$\boxed{LJ = 23}$$

(41) C

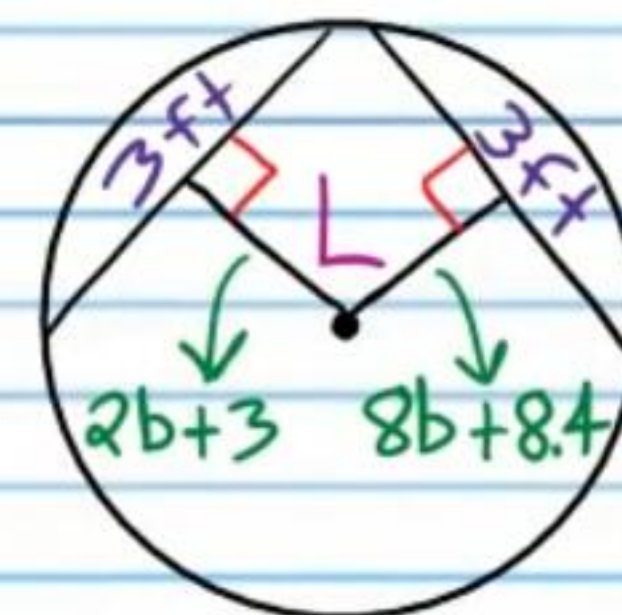


$$4c - 3 = 9c - 28$$

$$25 = 5c$$

$$\boxed{5 = c}$$

(40) b

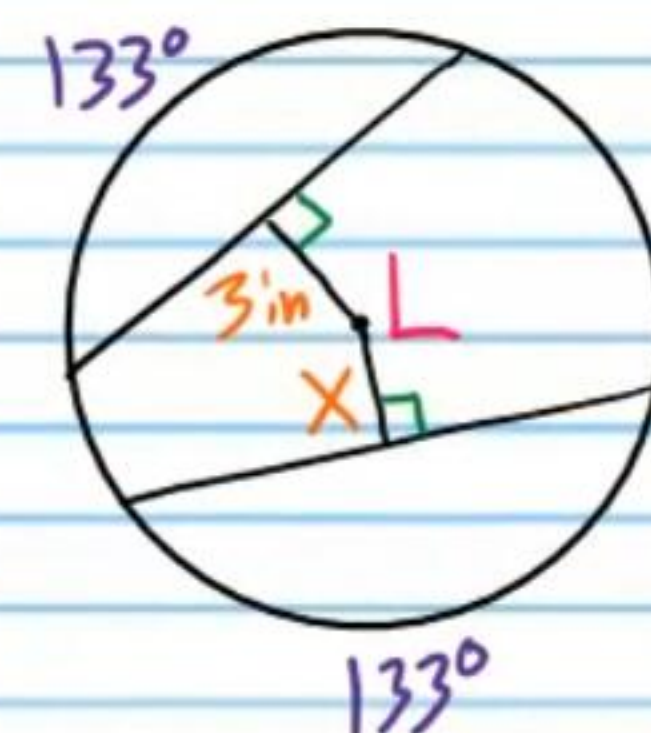


$$2b + 3 = 8b + 8.4$$

$$-5.4 = 6b$$

$$\boxed{-0.9 = b}$$

(42) X



$$\boxed{X = 3 \text{ in}}$$

Graphs of Circles

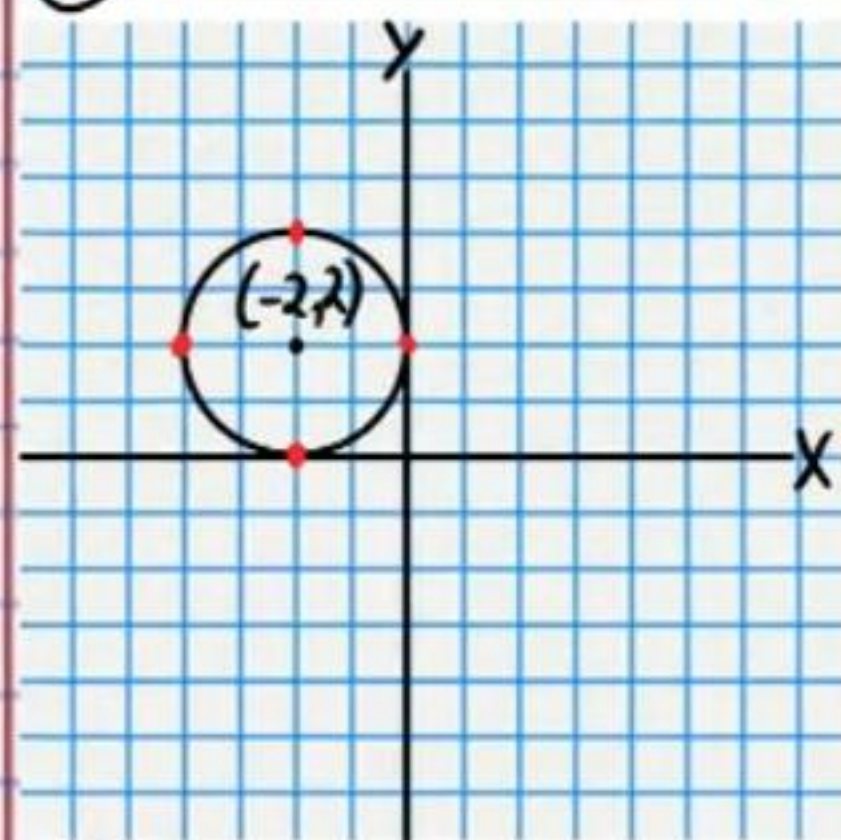
Equation of a Circle

$$(x-h)^2 + (y-k)^2 = r^2$$

With center (h,k) and radius r .

Write the equations of the following circles.

43

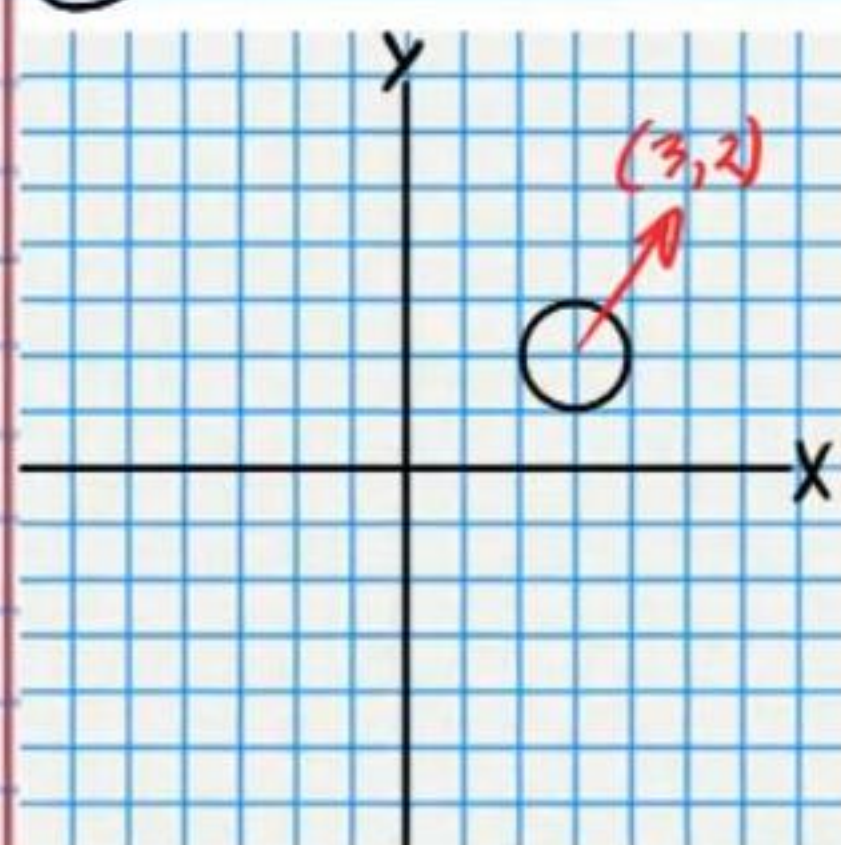


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-2))^2 + (y-2)^2 = 2^2$$

$$(x+2)^2 + (y-2)^2 = 4$$

44

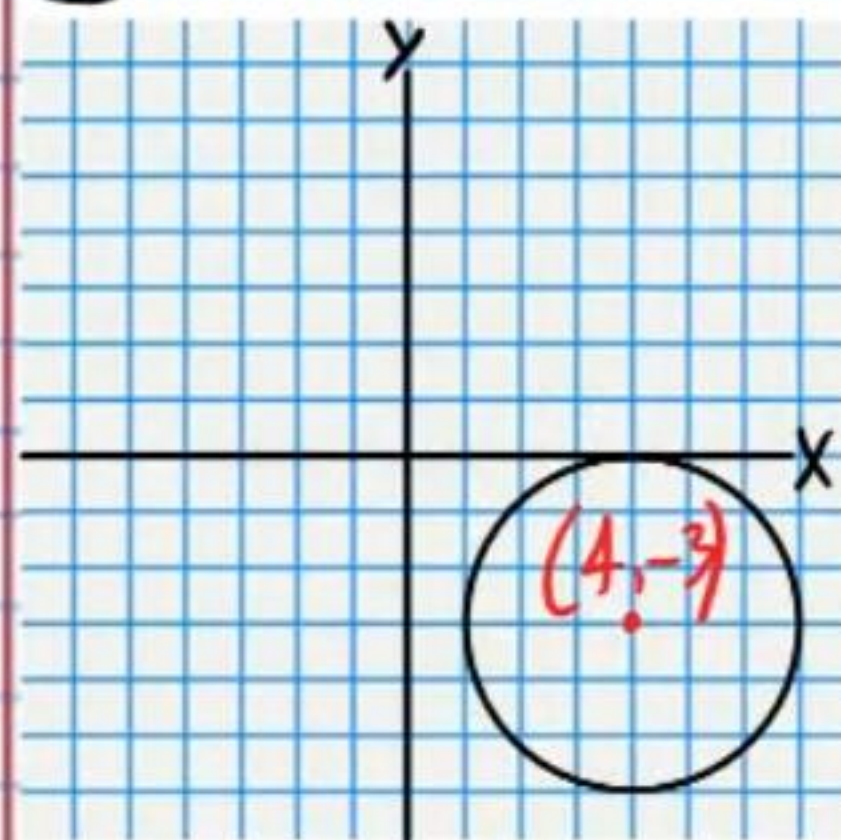


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-3)^2 + (y-2)^2 = 1^2$$

$$(x-3)^2 + (y-2)^2 = 1$$

45

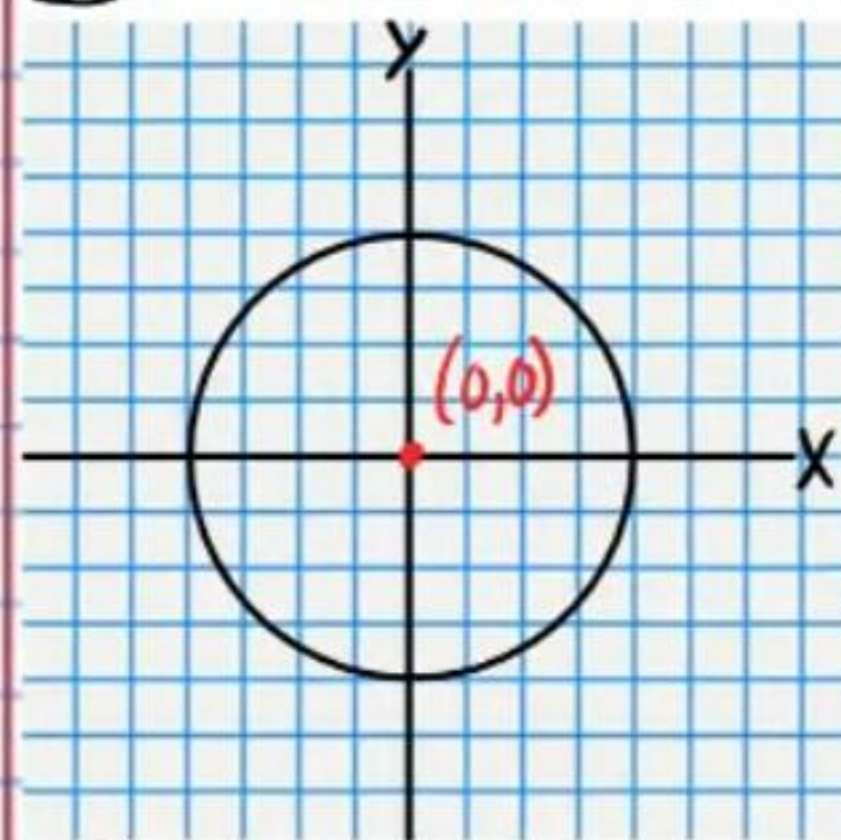


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-4)^2 + (y-(-3))^2 = 3^2$$

$$(x-4)^2 + (y+3)^2 = 9$$

46

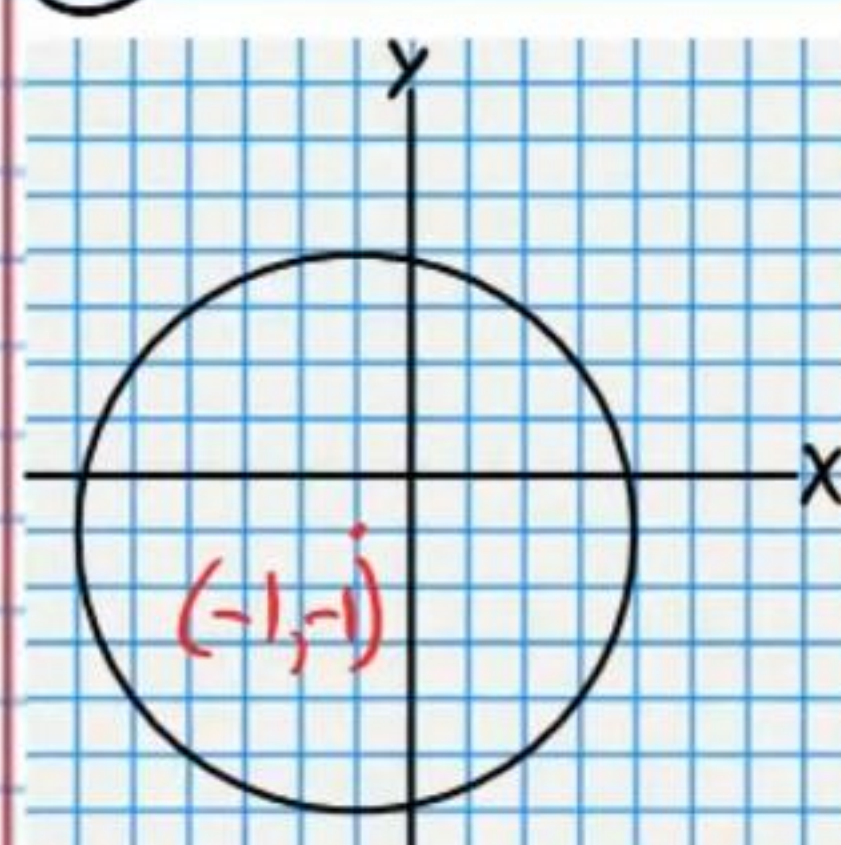


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-0)^2 + (y-0)^2 = 4^2$$

$$x^2 + y^2 = 16$$

47

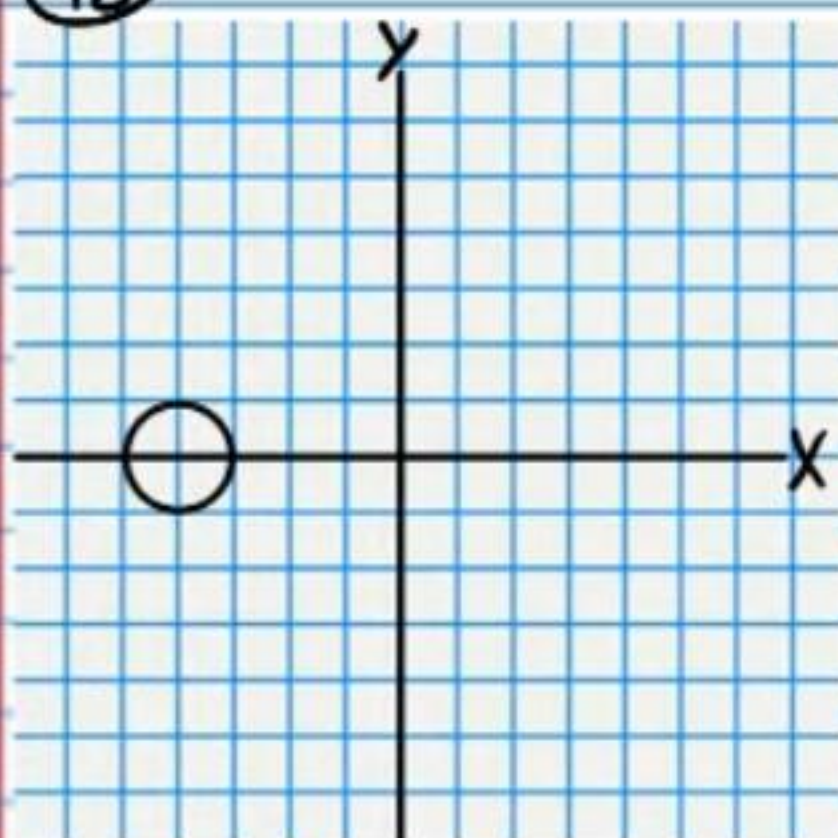


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-1))^2 + (y-(-1))^2 = 5^2$$

$$(x+1)^2 + (y+1)^2 = 25$$

48

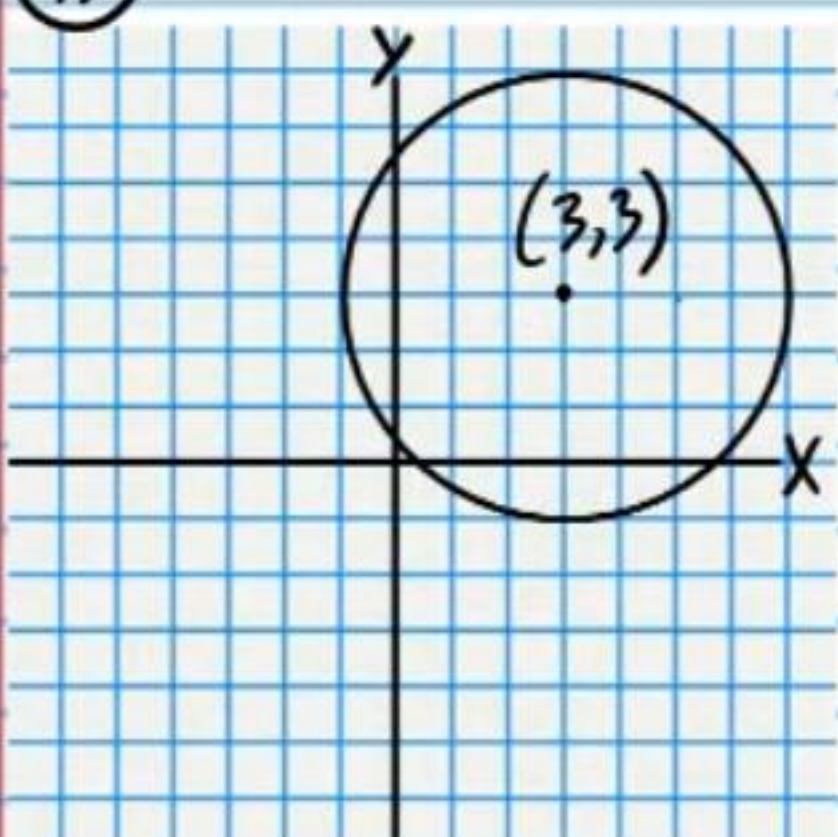


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-4))^2 + (y-0)^2 = 1^2$$

$$(x+4)^2 + y^2 = 1$$

49

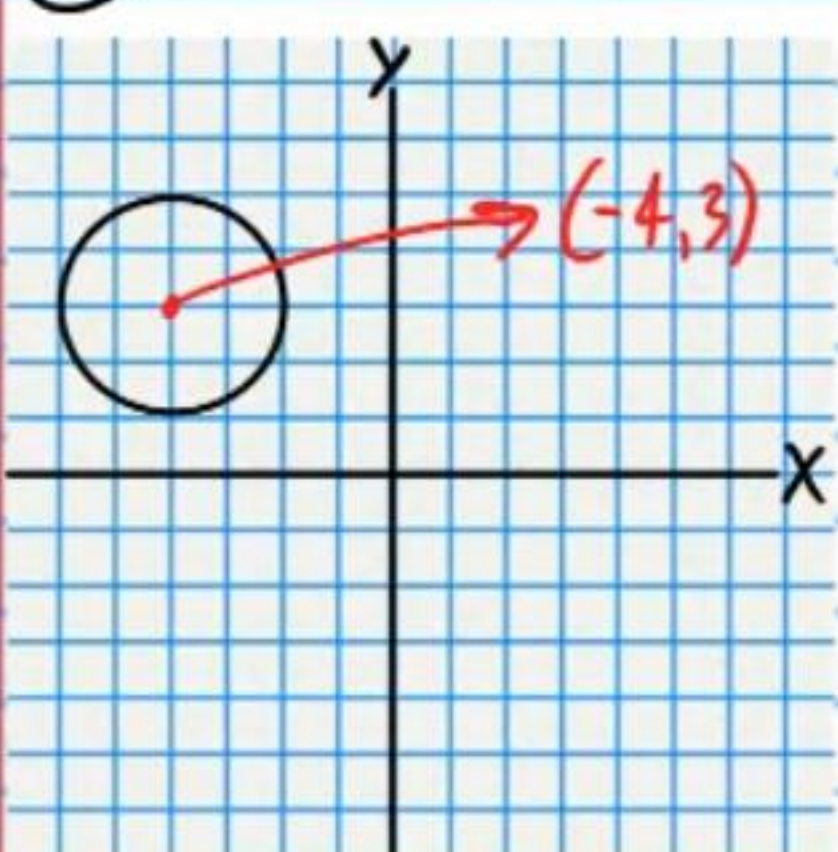


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-3)^2 + (y-3)^2 = 4^2$$

$$(x-3)^2 + (y-3)^2 = 16$$

50



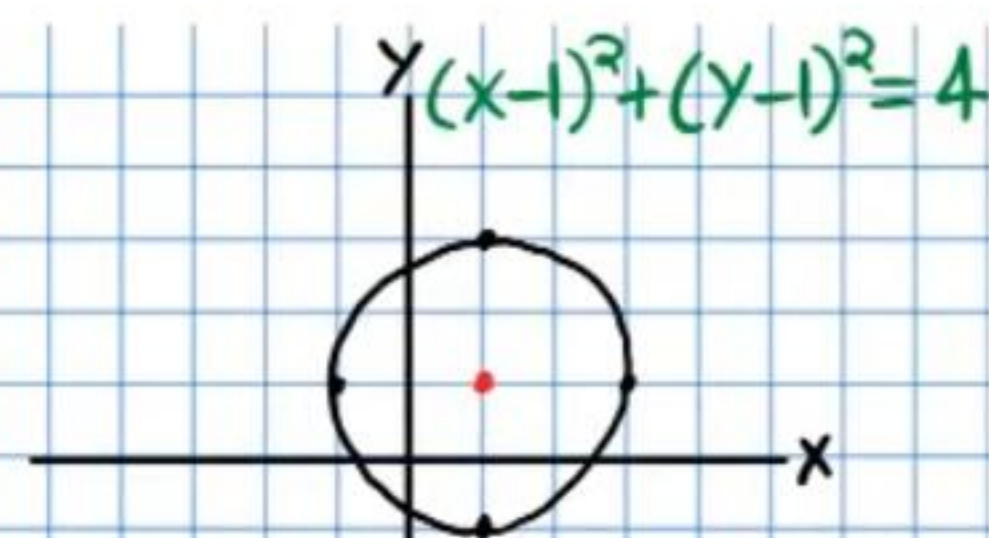
$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-4))^2 + (y-3)^2 = 2^2$$

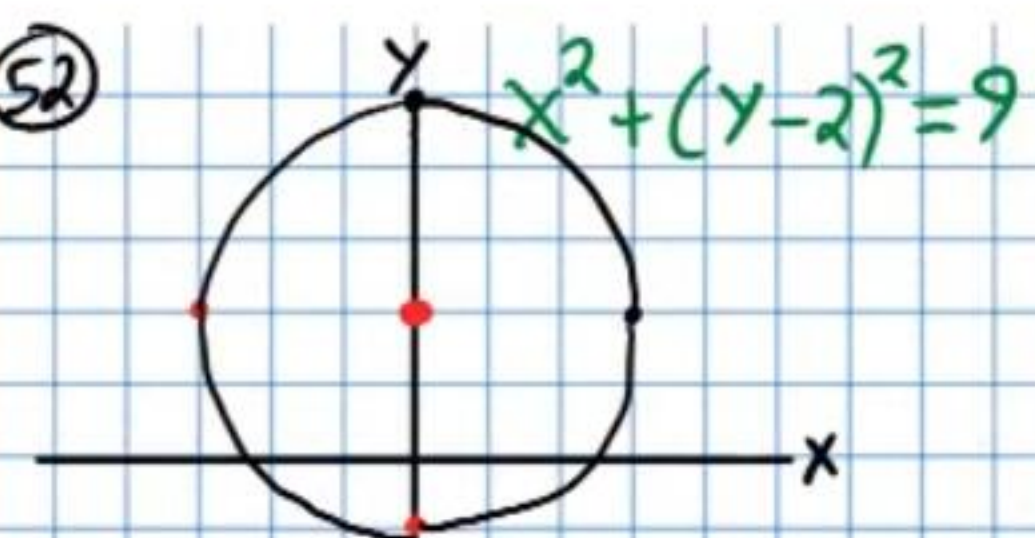
$$(x+4)^2 + (y-3)^2 = 4$$

Graph the equations below.

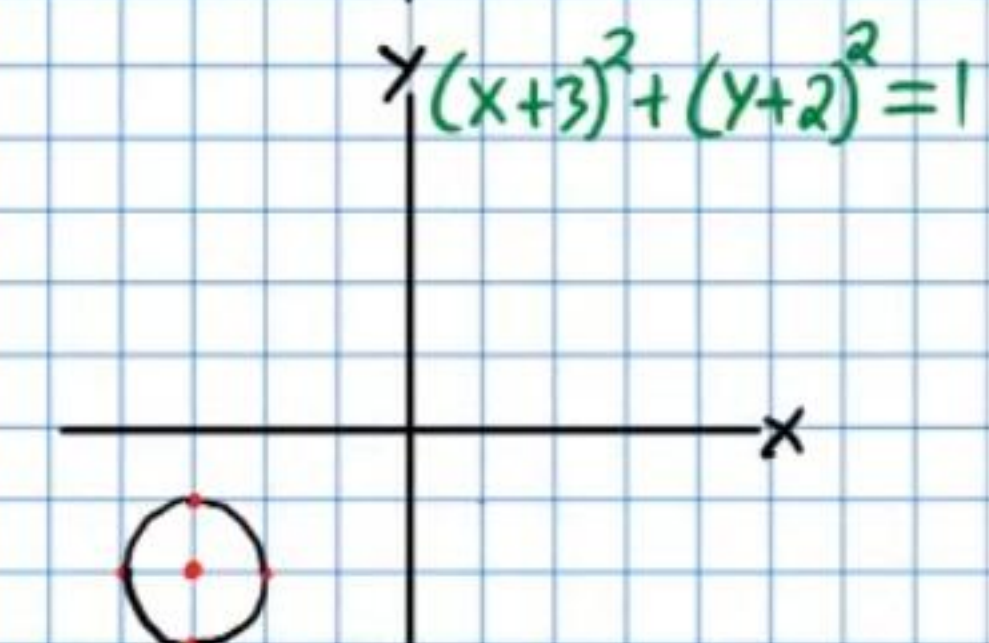
51



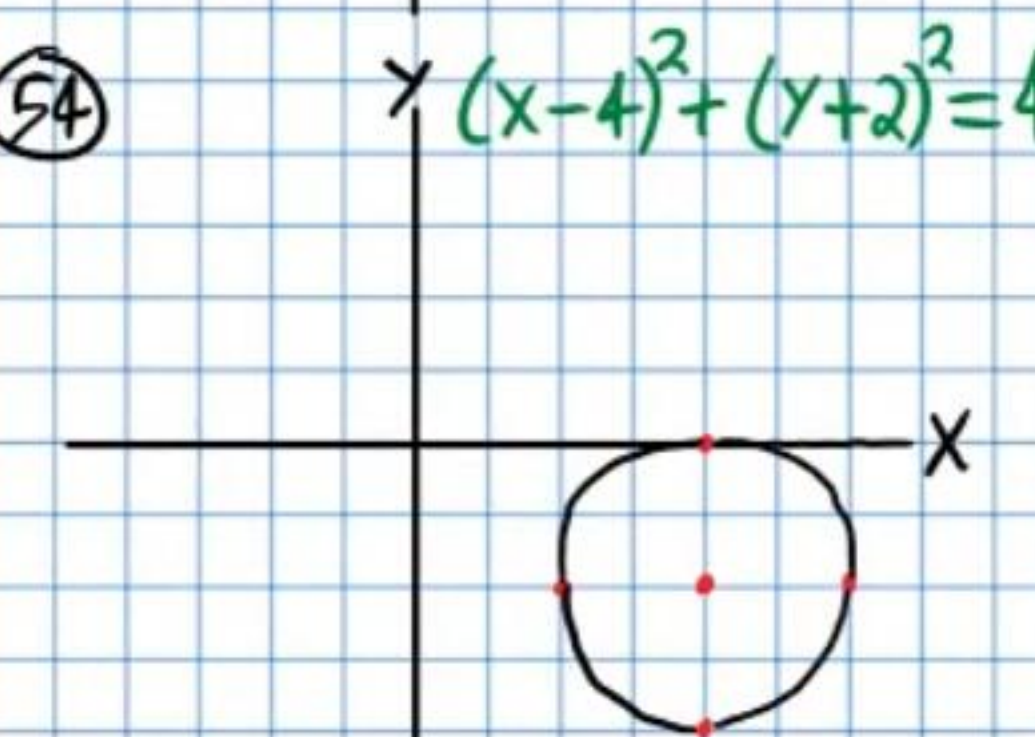
52



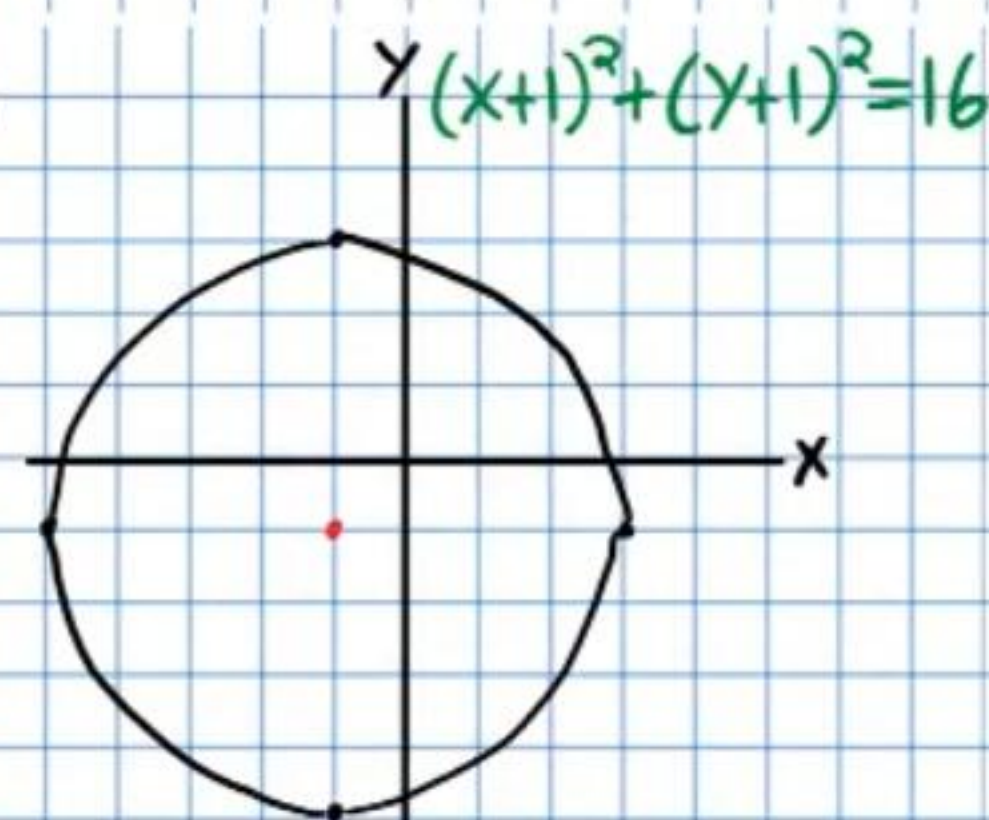
53



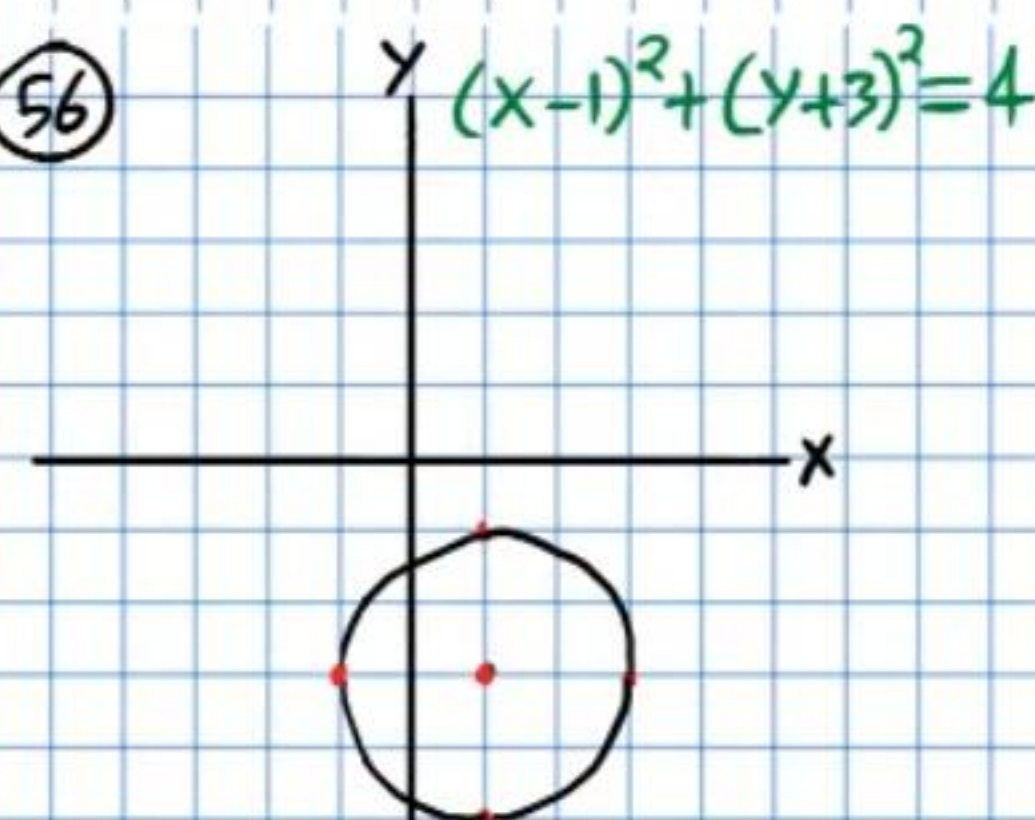
54



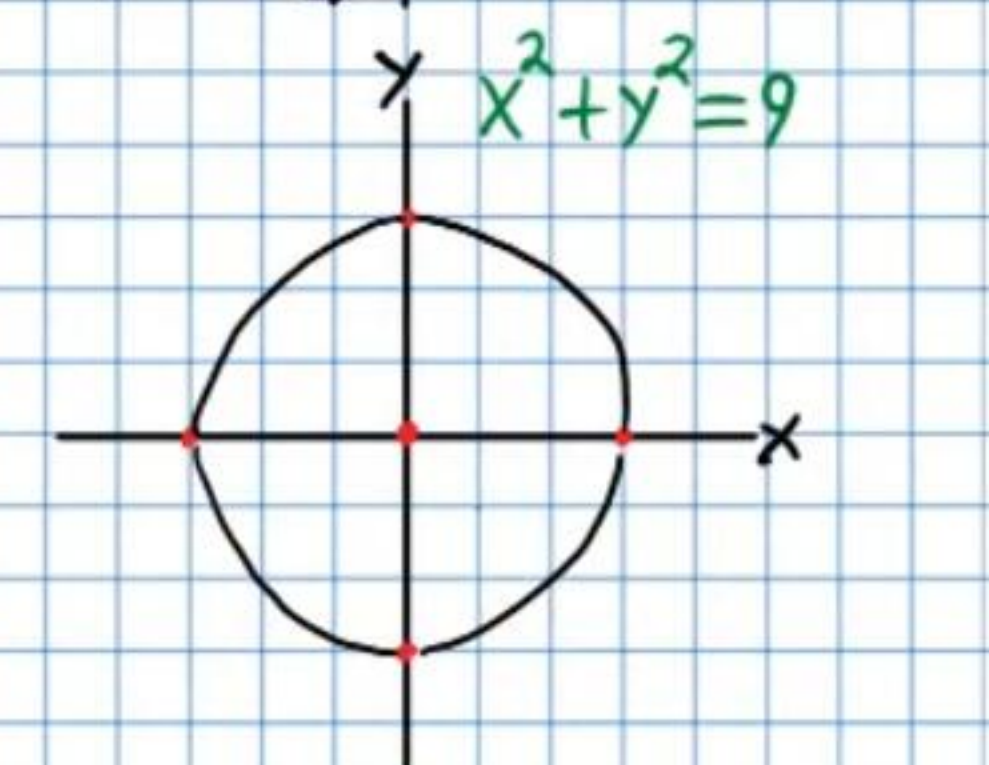
55



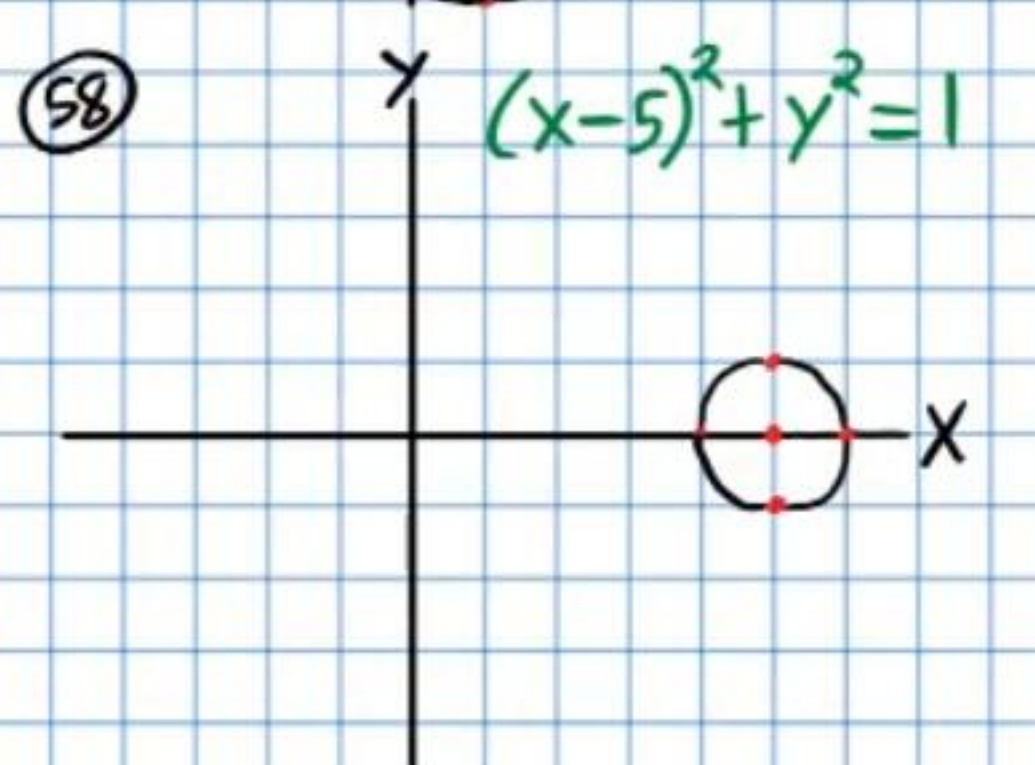
56



57



58



Write the equations of the circles with the given characteristics.

59) Center: $(-3, 5)$
Point on the circle: $(1, 8)$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(1 - (-3))^2 + (8 - 5)^2 = r^2$$

$$(1+3)^2 + 3^2 = r^2$$

$$4^2 + 3^2 = r^2$$

$$16 + 9 = r^2$$

$$25 = r^2$$

$$(x+3)^2 + (y-5)^2 = 25$$

60) Center: $(1, 7)$
Point on the circle: $(-2, 5)$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(-2-1)^2 + (5-7)^2 = r^2$$

$$(-3)^2 + (-2)^2 = r^2$$

$$9 + 4 = r^2$$

$$13 = r^2$$

$$(x-1)^2 + (y-7)^2 = 13$$

61) Center: $(2, -5)$
Point on the circle: $(1, 1)$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(1-2)^2 + (1-(-5))^2 = r^2$$

$$(-1)^2 + (1+5)^2 = r^2$$

$$1 + 6^2 = r^2$$

$$1 + 36 = r^2$$

$$37 = r^2$$

$$(x-2)^2 + (y+5)^2 = 37$$

62) Center: $(-5, 6)$
Point on the circle: $(-7, -2)$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(-7-(-5))^2 + (-2-6)^2 = r^2$$

$$(-7+5)^2 + (-8)^2 = r^2$$

$$(-2)^2 + 64 = r^2$$

$$4 + 64 = r^2$$

$$68 = r^2$$

$$(x+5)^2 + (y-6)^2 = 68$$

⑥③ Diameter with endpoints (4, 2) & (8, 6). ⑥④ Diameter with endpoints (5, -7) & (3, -4).

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{4 + 8}{2}, \frac{2 + 6}{2} \right)$$

$$\left(\frac{12}{2}, \frac{8}{2} \right)$$

(6, 4) → center

(4, 2) → point

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(4 - 6)^2 + (2 - 4)^2 = r^2$$

$$(-2)^2 + (-2)^2 = r^2$$

$$4 + 4 = r^2$$

$$8 = r^2$$

$$(x - 6)^2 + (y - 4)^2 = 8$$

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{5 + 3}{2}, \frac{-7 + (-4)}{2} \right)$$

$$\left(\frac{8}{2}, \frac{-11}{2} \right)$$

(4, -5.5) → center

(5, -7) → point

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(5 - 4)^2 + (-7 - (-5.5))^2 = r^2$$

$$1^2 + (-7 + 5.5)^2 = r^2$$

$$1 + (-1.5)^2 = r^2$$

$$1 + 2.25 = r^2$$

$$3.25 = r^2$$

$$(x - 4)^2 + (y + \frac{11}{2})^2 = 3.25$$

⑥⑤ Diameter with endpoints (6, -1) & (4, 1). ⑥⑥ Diameter with endpoints (-7, -3) & (-2, -4).

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{6 + 4}{2}, \frac{-1 + 1}{2} \right)$$

$$\left(\frac{10}{2}, \frac{0}{2} \right)$$

(5, 0) → center

(4, 1) → point

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(4 - 5)^2 + (1 - 0)^2 = r^2$$

$$(-1)^2 + 1^2 = r^2$$

$$1 + 1 = r^2$$

$$2 = r^2$$

$$(x - 5)^2 + (y - 0)^2 = 2$$

$$(x - 5)^2 + y^2 = 2$$

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{-7 + (-2)}{2}, \frac{-3 + (-4)}{2} \right)$$

$$\left(\frac{-9}{2}, \frac{-7}{2} \right)$$

(-4.5, -3.5) → center

(-7, -3) → point

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(-7 - (-4.5))^2 + (-3 - (-3.5))^2 = r^2$$

$$(-7 + 4.5)^2 + (-3 + 3.5)^2 = r^2$$

$$(-2.5)^2 + (0.5)^2 = r^2$$

$$6.25 + 0.25 = r^2$$

$$6.5 = r^2$$

$$\left(x + \frac{9}{2} \right)^2 + \left(x + \frac{7}{2} \right)^2 = 6.5$$